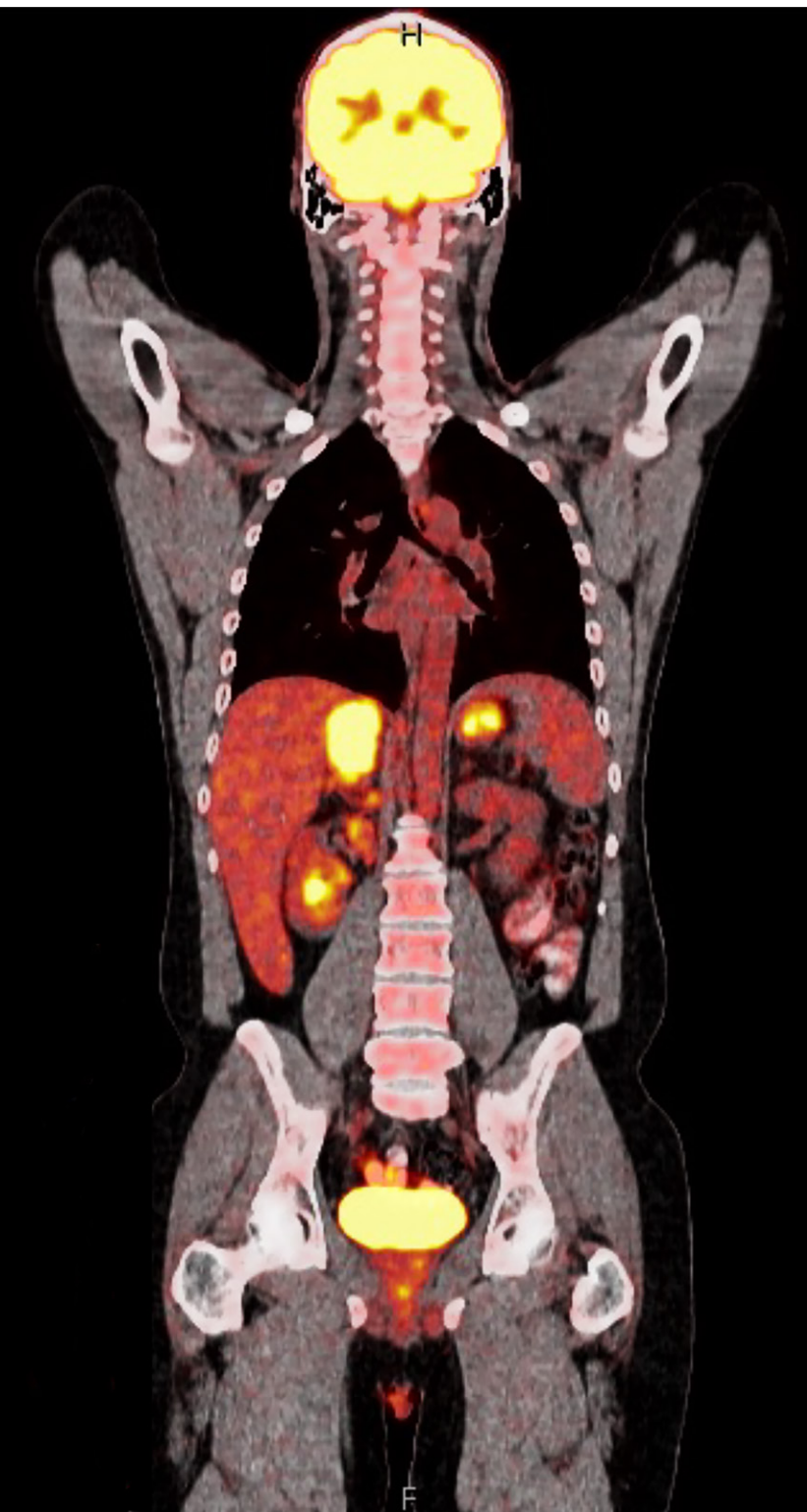


Imaging of Castleman Disease

Marika A. Pitot, MD • Ali D. Tahboub Amawi, MD • Lauren F. Alexander, MD • Jordan D. LeGout, MD • Lara Walkoff, MD
Patrick J. Navin, MB BCh BAO • Akira Kawashima, MD, PhD • Adam J. Wood, DO • Angela Dispenzieri, MD • Sudhakar K. Venkatesh, MD

Author affiliations, funding, and conflicts of interest are listed at [the end of this article](#).



Castleman disease (CD) is a group of rare and complex lymphoproliferative disorders that can manifest in two general forms: unicentric CD (UCD) and multicentric CD (MCD). These two forms differ in clinical manifestation, imaging appearances, treatment options, and prognosis. UCD typically manifests as a solitary enlarging mass that is discovered incidentally or after development of compression-type symptoms. MCD usually manifests acutely with systemic symptoms including fever and weight loss. As a whole, CD involves lymph nodes throughout the chest, neck, abdomen, pelvis, and axilla and can have a wide variety of imaging appearances. Most commonly, lymph nodes or masses in UCD occur in the chest, classically with well-defined borders, hyperenhancement, and possible characteristic patterns of calcification and/or feeding vessels. Lymph nodes affected by MCD, while also hyperenhancing, tend to involve multiple nodal chains and manifest alongside anasarca or hepatosplenomegaly. The *polyneuropathy, organomegaly, endocrinopathy, monoclonal plasma cell disorder, and skin changes* (POEMS) subtype of MCD may demonstrate lytic or sclerotic osseous lesions in addition to features typical of MCD. Since a diagnosis of CD based solely on imaging findings is often not possible, pathologic confirmation with core needle biopsy and/or surgical excision is necessary. Nevertheless, imaging plays a crucial role in supporting the diagnosis of CD, guiding appropriate regions for biopsy, and excluding other potential causes or mimics of disease. CT is frequently the initial imaging technique used in evaluating potential CD. MRI and PET play important roles in thoroughly evaluating the disease and determining its extent, especially the MCD form. Complete surgical excision is typically curative for UCD. MCD usually requires systemic therapy.

©RSNA, 2023 • radiographics.rsna.org

Supplemental
Material



Quiz questions for this article are available in the [supplemental material](#).

RadioGraphics 2023; 43(8):e220210
<https://doi.org/10.1148/rg.220210>

Content Codes: CH, CT, GI, GU, MR, NM

Abbreviations: CD = Castleman disease; EBV = Epstein-Barr virus; FDC = follicular dendritic cell; FDG = fluorodeoxyglucose; HHV8 = human herpesvirus 8; HIV = human immunodeficiency virus; HVV = hyaline vascular variant; iMCD = idiopathic MCD; MCD = multicentric CD; PCV = plasma cell variant; POEMS = polyneuropathy, organomegaly, endocrinopathy, monoclonal plasma cell disorder, and skin changes; TAFRO = thrombocytopenia, ascites, reticulin fibrosis, renal dysfunction, and organomegaly; UCD = unicentric CD

TEACHING POINTS

- Castleman disease (CD), also known as angiofollicular lymph node hyperplasia, is a complex lymphoproliferative disorder broadly divided into two forms: unicentric CD (UCD) and multicentric CD (MCD). The disease likely encompasses at least four different entities with overlapping histopathologic features but distinct causes, clinical manifestations, radiologic features, treatment approaches, and outcomes.
- Most commonly, UCD manifests as a solitary enlarging mass, which can be discovered incidentally at imaging performed for other reasons, be palpated at physical examination, or be discovered owing to compression-type symptoms on adjacent structures.
- CD most commonly manifests as unicentric disease with various sites of involvement. The chest is the most common site (30%–70%), followed by the neck (10%–40%), abdomen and pelvis (12%–39%), and axilla (4%–5%).
- Multiple modalities may demonstrate prominent feeding vessels to the lymph node or mass, which can affect the surgical approach.
- Lymphoma is a common mimic that is often a diagnostic consideration for presumed CD in different parts of the body. Lymphoma can be confused with CD in the mediastinum, MCD, or nonspecific masses in other parts of the body.

Introduction

Castleman disease (CD), also known as angiofollicular lymph node hyperplasia, is a complex lymphoproliferative disorder broadly divided into two forms: unicentric CD (UCD) and multicentric CD (MCD). The disease likely encompasses at least four different entities with overlapping histopathologic features but distinct causes, clinical manifestations, radiologic features, treatment approaches, and outcomes (1). CD was first described by Benjamin Castleman in 1954, and our understanding of the disease and its many facets has continued to evolve since that time (2,3).

CD most commonly affects lymph nodes, arising in nodal stations throughout the body but most commonly involving the chest (4). MCD manifesting with systemic symptoms and lymphadenopathy can mimic lymphoma, whereas UCD manifesting as a single mass in the mediastinum can mimic a neurogenic tumor. Therefore, diagnosis requires a combination of clinical, radiologic, and pathologic information. Understanding the imaging features of CD allows accurate and timely diagnosis and enables subsequent treatment.

The purpose of this article is to provide an overview of the classification, epidemiology, pathogenesis, clinical features, and histopathologic variants of CD as well as to review the radiologic features of CD in various parts of the body and across different imaging modalities. Potential mimics, in particular lymphoma, are discussed. Finally, the article covers the treatment options and posttreatment response, as well as impor-

Table 1: Broad Distinctions between Unicentric and Multicentric Forms of CD

UCD	MCD
Localized to a single node or nodal region	Diffuse systemic involvement
Asymptomatic	Constitutional symptoms such as fever and myalgia
Indolent	Acute onset
Incidentally discovered	Widespread lymphadenopathy, hepatosplenomegaly, anasarca
Often curable surgically	Requires systemic therapy

tant prognostic factors. Representative cases are presented with key imaging findings highlighted.

Classification

The clinical manifestation of CD is broadly divided into UCD and MCD (Table 1). MCD encompasses three separate subtypes: idiopathic MCD (iMCD), human herpesvirus 8 (HHV8)–associated MCD (HHV8-MCD), and polyneuropathy, organomegaly, endocrinopathy, monoclonal plasma cell disorder, and skin changes (POEMS) syndrome–associated MCD (POEMS-MCD). The iMCD subtype can be further divided into iMCD with thrombocytopenia, ascites, reticulin fibrosis, renal dysfunction, and organomegaly (iMCD-TAFRO) and iMCD not otherwise specified (iMCD-NOS) (1,5). The most well-defined and understood variant of MCD is HHV8-MCD (6).

Epidemiology

Epidemiologic data on CD are limited and vary greatly, partly owing to the previous absence of a specific International Classification of Diseases (ICD) code before 2016 and lack of evidence-based consensus diagnostic criteria until 2017. UCD and MCD differ in terms of incidence, populations affected, and presence of identified risk factors.

Unicentric CD

Of the approximately 6500–7700 cases of CD that occur annually in the United States, 75% are UCD (5). The estimated incidence of UCD is 16 per million person-years with a median age of UCD onset in the 4th decade (range, 2–84 years) (5,6). There is no clear gender preference or known risk factors for UCD, regardless of ethnicity, genetic predisposition, or family history (5,6).

Multicentric CD

The estimated incidence of HHV8-MCD, iMCD, and POEMS-MCD is five per million person-years each (6). The estimated proportion of POEMS in patients with MCD ranges from 13.1% to 32% (6). The incidence of TAFRO in patients with iMCD is estimated to range from 2% to 21% (6). The median age of onset for MCD is in the 5th to 7th decades (range, 1–83 years) with no consistent gender preference (5,6).

Unlike UCD, MCD has several identified risk factors, which differ on the basis of subtype, including viral infections, plasma cell dyscrasias, and immune dysfunction. Although

HHV8-MCD is its own clinicopathologic entity regardless of human immunodeficiency virus (HIV) status, HIV infection poses a significant risk owing to associated immunocompromise, enabling dormant forms of HHV8 to escape immune control (6–8). The largest immunocompromised population with HHV8-MCD is therefore HIV positive (6). Other risk factors for HHV8-MCD related to immunodeficiency include age-related immune senescence and organ transplantation (6,7).

The seroprevalence of HHV8 varies geographically from as low as 2%–4% in Northern Europe, Southeast Asia, and the Caribbean to up to 40% in regions of sub-Saharan Africa. Increased risk of HHV8-MCD has also been associated with no previous highly active antiretroviral therapy (HAART) exposure, non-White ethnicity, and a nadir CD4 cell count greater than 200 per cubic millimeter (6). Surprisingly, the incidence of HHV8-MCD has increased in the HAART era, possibly owing to immune reconstitution (6,9). It has been suggested that in the setting of decreased viral loads after treatment, patients with HIV infection who are coinfecting with HHV8 subsequently are able to mount a stronger immune response to their HHV8 coinfection, leading to elevated inflammatory markers and constitutional symptoms, prompting workup and ultimate diagnosis of HHV8-MCD (10).

Clinical Features

Clinical manifestations of CD are variable and dependent on the clinicopathologic subtype and any coexistent or associated systemic disorders, infections, or complications.

Unicentric CD

Most commonly, UCD manifests as a solitary enlarging mass, which can be discovered incidentally at imaging performed for other reasons, be palpated at physical examination, or be discovered owing to compression-type symptoms on adjacent structures.

Compressive symptoms of UCD confined to the chest can include cough, hemoptysis, dyspnea, or chest pain; compressive symptoms related to abdominal, retroperitoneal, or pelvic disease include abdominal or back pain and rarely bowel or ureteral obstruction. Disease associated with peripheral lymph node chains such as the neck, axilla, or inguinal region may manifest as palpable nontender adenopathy (7).

Multicentric CD

MCD typically manifests acutely with systemic symptoms including fever, night sweats, and weight loss. Anasarca and fluid retention may exist as lower extremity edema, pleural or pericardial effusion, or ascites. Hepatosplenomegaly or generalized lymphadenopathy are also common (7).

Laboratory tests in patients with MCD may reveal anemia, hypoalbuminemia, or other cytopenias, as well as elevated levels of inflammatory markers and cytokines, including interleukin 6 (IL-6) and interleukin 10 (IL-10) (5,7).

Associations and Complications

The various CD subtypes have been associated with several other disease entities and syndromes of prognostic and therapeutic importance.

TAFRO-MCD

The iMCD-TAFRO variant is defined as MCD but with thrombocytopenia, anasarca, fever, bone marrow reticulin fibrosis (or renal insufficiency), and organomegaly (hepatosplenomegaly and lymphadenopathy).

POEMS-MCD

Concurrence of POEMS syndrome with MCD (POEMS-MCD) is an important consideration in patients with CD and is part of the diagnostic algorithm for workup. POEMS-MCD is a paraneoplastic syndrome caused by an underlying plasma cell neoplasm. Up to 30% of patients with POEMS syndrome may also have MCD (11). Patients with POEMS-MCD will have peripheral polyneuropathy as a defining feature, along with MCD symptoms.

In addition to polyneuropathy, other features include organomegaly, endocrinopathy, monoclonal plasma cells, and a variety of skin changes including hyperpigmentation, vascular skin changes such as Raynaud phenomenon, hypertrichosis, and clubbing. POEMS syndrome can manifest with sclerotic bone lesions, papilledema, extravascular volume overload, thrombocytosis, elevated vascular endothelial growth factor (VEGF) level, and abnormal pulmonary function (6,12).

POEMS syndrome is differentiated from multiple myeloma by several features, including the predominant symptoms of neuropathy, endocrine dysfunction, and volume overload without significant bone pain, extensive bone marrow infiltration by plasma cells, or renal failure. POEMS syndrome is also characterized by a higher frequency of sclerotic bone lesions and a better overall survival rate compared with those of multiple myeloma (13). Up to 50% of cases of POEMS syndrome coincide with sclerotic myeloma (11).

Lymphoma

Lymphoma should be considered in the differential diagnosis of CD (14). Patients with UCD or MCD are also at increased risk of developing lymphoma (5,7). Up to 25% of MCD cases have been associated with development of non-Hodgkin lymphoma (NHL), most commonly B-cell subtypes (11,15), although Hodgkin lymphoma and plasmacytoma have also been described.

Patients with HHV8-MCD also have a unique risk of developing primary effusion lymphoma (PEL), a rare form of NHL, which is also associated with Epstein-Barr virus (EBV) infection (5,11,14). PEL is characterized by malignant fluid accumulation in the pleural, pericardial, and peritoneal spaces without lymphadenopathy (16). Plasmablastic lymphomas arising out of HHV8-MCD are not associated with EBV but are thought to be secondary to a hyperproliferative state induced by HHV8 in the infected B immunoblasts (17).

In patients with CD, simultaneous or sequential development of lymphoma may be indicated by new laboratory abnormalities such as anemia or the development of B symptoms. Certain imaging features including increasing adenopathy, ascites, or new splenomegaly may also support the diagnosis (14). However, owing to similarities in both the clinical and imaging appearance, definitive diagnosis typically requires a tissue biopsy (14).

Follicular Dendritic Cell Sarcoma

Follicular dendritic cell (FDC) sarcoma is a rare neoplasm arising from the FDCs of germinal centers in lymph nodes. Although UCD has been known to be a predisposing factor, only a few cases of CD transforming into FDC sarcoma have been documented in the literature (18). FDC sarcoma most commonly involves cervical, axillary, and mediastinal lymph node chains but can also occur in extranodal sites, including those in the head and neck region as well as the abdomen.

No specific imaging features have been identified that would suggest transformation of CD into FDC sarcoma (19). However, one case report described FDC sarcoma within a small hypoenhancing mass near the site of prior biopsy-proven CD (20). The clinical presentation of patients with FDC sarcoma can also parallel that of CD, related to mass effect of involved sites, including chest pain, cough, abdominal pain, or abdominal distention. Patients with FDC sarcoma can develop fever as well as elevated white blood cell (WBC) count and inflammatory markers such as C-reactive protein (CRP).

Kaposi Sarcoma

Owing to shared HHV8-mediated pathogenesis, HHV8-MCD may also manifest with other malignancies such as Kaposi sarcoma. The frequency of concurrent diagnosis of Kaposi sarcoma is reported to be 72% in cases of HIV-related MCD and less common in those with MCD who are negative for HIV (7,15). Kaposi sarcoma is a low- to intermediate-grade mesenchymal neoplasm that involves the lymphatic and vascular systems. The imaging features of Kaposi sarcoma vary greatly depending on the variant and affected organ system but typically involve the skin, mucosal membranes, lungs, liver, and gastrointestinal tract (21).

Radiologic manifestations in the chest include characteristic flame-shaped peribronchovascular soft-tissue densities and ground-glass halos in the parenchyma. Other manifestations include pleural effusions and mediastinal or hilar lymphadenopathy (21,22). Multifocal hypoenhancing nodules in the liver and spleen that enhance in the delayed phase may be seen, unlike in CD, where there is only hepatosplenomegaly. Bull's-eye or target lesions in the mucosa of the stomach may be seen in double-contrast barium studies (21).

A more specific discussion of the clinical symptoms and imaging features of Kaposi sarcoma is outside the scope of this article. In resource-poor areas, diagnosis of Kaposi sarcoma can be made by visually inspecting the characteristic violaceous skin lesions, although biopsy can confirm the diagnosis.

Other Associations

Patients with UCD have an increased risk of developing paraneoplastic pemphigus, a chronic and debilitating blistering mucocutaneous disease (7,11). iMCD has been associated with a range of disorders, which in addition to paraneoplastic pemphigus include cryptogenic organizing pneumonia (COP), autoimmune cytopenias, polyneuropathy (without meeting the criteria for POEMS syndrome), glomerulonephropathy, and inflammatory myofibroblastic tumor (23). HHV8-MCD may also manifest with hemophagocytic syndrome (11).

Pathogenesis

The exact cause of CD is still not fully understood, but it is believed to be related to overproduction of interleukin 6 (IL-6) (5). IL-6 is a glycoprotein cytokine produced in various cells including immune cells, hemopoietic cells, and epithelial cells (24). T cells, B cells, and macrophages are known to secrete IL-6, which acts on multiple targets to induce secretion of acute phase proteins, stimulate cell growth and differentiation, regulate immune responses, support hemopoiesis, and activate the central nervous system. In vivo studies have shown that activation of IL-6 can lead to fever, fatigue, anorexia, and lymphadenopathy as well as laboratory abnormalities, including increased acute phase reactants (C-reactive protein [CRP], serum amyloid A [SAA], fibrinogen, and hepcidin), renal dysfunction, anemia, osteoporosis, thrombocytosis, and hypoalbuminemia (24).

The role of IL-6 in UCD is less clear, especially since most cases of UCD do not manifest with systemic inflammatory symptoms. Although various theories—including viral, neoplastic or clonal, and inflammatory mechanisms—have been proposed, UCD is now believed to have a neoplastic origin (25). This theory has been supported by notable histopathologic similarities between affected tissue and stromal cells or FDCs and the development of FDC sarcoma in some cases of UCD (6).

The role of IL-6 in MCD pathogenesis has been supported by studies that show a correlation between high levels of IL-6 and clinical symptoms. In these same patients, IL-6 production in affected lymph nodes has been confirmed through pathologic examination (26). Additionally, anti-IL-6 therapies have been effective in treating MCD, further supporting its role in the disease's pathogenesis (24,26). HHV8 is a well-established cause of HHV8-MCD. During periods of immunodeficiency, latent HHV8 can replicate in lymph nodes and signal the release of cytokines including IL-6, leading to clinical and pathologic symptoms (25).

The pathogenesis of POEMS-MCD is also thought to be related to cytokine overproduction, specifically due to genomically modified monoclonal plasma cells that produce elevated levels of vascular endothelial growth factor (VEGF), IL-6, and IL-12 (25).

The cause of iMCD, including iMCD-TAFRO, is also not well understood, although IL-6 is likely to be involved (27). As the features of this subtype are more heterogeneous than those of others, it is believed that a combination of factors may contribute to the development of iMCD, including chronic inflammation, lymphoid hamartomatous hyperplasia, viral infections other than HHV8, paraneoplastic or autoimmune phenomena, and immunodeficiency (5,7,14,25,28). A genetic predisposition to iMCD has been suggested by an increased prevalence of iMCD in Polynesians living in New Zealand (6). However, the definite designation for iMCD on the spectrum of autoimmune, malignant, and infectious diseases remains unknown and may even vary between patients (23).

Histopathologic Subtypes and Findings

The two major histopathologic subtypes of CD are hyaline vascular variant (HV) and plasma cell variant (PCV). Cases

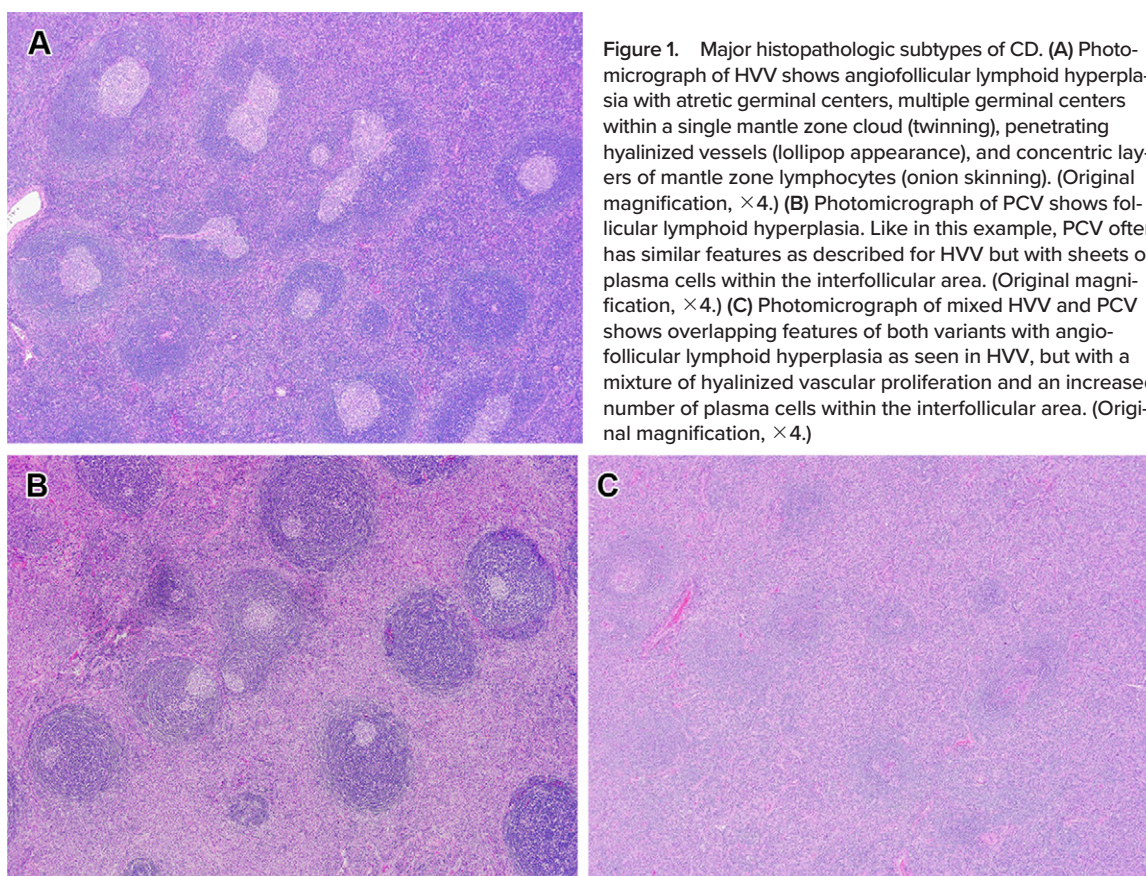


Figure 1. Major histopathologic subtypes of CD. (A) Photomicrograph of HVV shows angiofollicular lymphoid hyperplasia with atretic germinal centers, multiple germinal centers within a single mantle zone cloud (twinning), penetrating hyalinized vessels (lollipop appearance), and concentric layers of mantle zone lymphocytes (onion skinning). (Original magnification, $\times 4$.) (B) Photomicrograph of PCV shows follicular lymphoid hyperplasia. Like in this example, PCV often has similar features as described for HVV but with sheets of plasma cells within the interfollicular area. (Original magnification, $\times 4$.) (C) Photomicrograph of mixed HVV and PCV shows overlapping features of both variants with angiofollicular lymphoid hyperplasia as seen in HVV, but with a mixture of hyalinized vascular proliferation and an increased number of plasma cells within the interfollicular area. (Original magnification, $\times 4$.)

that demonstrate overlapping features are termed *mixed type* (Fig 1). Previously, HVV and PCV were thought to correspond to the UCD and MCD subtypes, respectively, but it is now recognized that UCD and MCD can be of any histopathologic variant (1,23,25).

Characteristic findings in HVV include interfollicular vascular proliferation with hyalinized walls, large follicles with regressed germinal centers, multiple germinal centers within a single mantle zone (twinning), penetrating hyalinized vessels (lollipop appearance), and concentric layers of mantle zone lymphocytes (onion skinning). The PCV type is characterized by sheets of plasma cells in the interfollicular areas. This variant also has large follicles that typically demonstrate hyperplastic germinal centers but may focally show the same regressive changes of HVV. The mixed variant shows a mixture of features of both HVV and PCV.

By definition, HHV8-MCD requires infection with HHV8, a rhadinovirus, which can be detected through immunohistochemical staining of tissue for latency-associated nuclear antigen (LANA)-1 (1,23). HHV8-MCD typically shows histologic findings of PCV and may also demonstrate lymphocyte depletion (especially in HIV-positive patients), a blurred border between the mantle zone lymphocytes and interfollicular area, and HHV8-positive plasmablasts situated in the mantle zones.

The histopathologic findings of CD are not pathognomonic on their own. Similar histologic findings can be seen in patients with autoimmune disorders, infections, and inflammatory conditions, and secondary causes must always be excluded. Clinical and laboratory findings are therefore

necessary to support a histologic diagnosis of CD (1). Additionally, various hemopoietic neoplasms can mimic HVV or PCV, including classic Hodgkin lymphoma, follicular lymphoma, angioimmunoblastic T-cell lymphoma, plasma cell neoplasms, and FDC sarcoma. If morphologic results warrant, these entities are excluded through an informed immunophenotypic workup.

Workup and Diagnosis

The diagnosis of CD is often suspected initially owing to radiographic findings (15). While imaging can provide clues to the diagnosis of CD and allow exclusion of possible mimics, definitive diagnosis of any of the four subtypes of CD requires tissue sampling of affected lymph nodes via biopsy, ideally excisional. If excision is not possible, a core needle biopsy is preferred over fine-needle aspiration (FNA), as FNA is not reliable for pathologic analysis of both UCD and MCD (7).

The first step in the diagnostic process is to determine the site of involvement, with single-site involvement suggesting UCD and multiple-site involvement suggesting MCD.

If there are multiple sites involved, patients should be evaluated for HHV8 infection and POEMS syndrome. Evaluation for HHV8 infection is made through immunohistochemical staining, while diagnosis of POEMS syndrome requires meeting separate mandatory major and minor criteria as defined by the Dispenzieri criteria (13), from which a positive result would suggest HHV8-MCD or POEMS-MCD, respectively. Of note, it is possible to have a diagnosis of POEMS syndrome without associated CD.

Table 2: Consensus Diagnostic Criteria for iMCD

Type of Criteria	Description
Major criteria (need both)	Histopathologic lymph node features consistent with iMCD spectrum Enlarged lymph nodes (≥ 1 cm in short-axis diameter) at two or more lymph node stations
Minor criteria (need at least two of 11)	Laboratory criteria Elevated CRP level or ESR Anemia Thrombocytopenia Hypoalbuminemia Renal dysfunction Polyclonal hypergammaglobulinemia Clinical criteria Constitutional symptoms: night sweats, fever, weight loss, or fatigue Large spleen or liver Fluid accumulation: edema, anasarca, ascites, or pleural effusion Eruptive cherry hemangiomas or violaceous papules Lymphocytic interstitial pneumonitis
Exclusion criteria (must rule out diseases that can mimic iMCD)	Infection-related disorders (HHV8, clinical EBV-lymphoproliferative disorders such as infectious mononucleosis or chronic active EBV, other uncontrolled infections causing widespread inflammation and adenopathy such as CMV, toxoplasmosis, HIV, active tuberculosis) Autoimmune or autoinflammatory diseases (SLE, RA, adult-onset Still disease, JIA, autoimmune lymphoproliferative syndrome) Malignant or lymphoproliferative disorders (lymphoma, multiple myeloma, primary lymph node plasmacytoma, FDC sarcoma, POEMS syndrome)

Note.—Adapted and reprinted, with permission, from reference 23. CMV = cytomegalovirus, CRP = C-reactive protein, ESR = erythrocyte sedimentation rate, JIA = juvenile idiopathic arthritis, RA = rheumatoid arthritis, SLE = systemic lupus erythematosus.

Diagnosis and classification of patients with suspected CD is based on fulfillment of major and minor criteria. Both major criteria (satisfactory histopathologic features and enlarged lymph nodes [≥ 1 cm in the short axis] at two or more sites) and at least two of 11 possible minor criteria, one of which must be a laboratory value, must be met for a diagnosis of CD to be made (Table 2).

Potential mimics of UCD include neurogenic tumors such as neurofibroma, schwannoma, and paraganglioma, which require histologic confirmation for exclusion. Other potential mimics of MCD must be excluded—including EBV-related disorders such as infectious mononucleosis or chronic active EBV, and inflammation and adenopathy caused by other infectious entities such as cytomegalovirus (CMV), HIV, and active tuberculosis—with appropriate serologic tests. Autoimmune causes such as rheumatoid arthritis, systemic lupus erythematosus (SLE), adult-onset Still disease, and juvenile idiopathic arthritis (JIA) should be excluded by clinical features and autoimmune serologic markers.

Sarcoidosis should be differentiated from MCD by tissue diagnosis. Lymphoproliferative disorders such as lymphoma, multiple myeloma, plasmacytoma, and FDC sarcoma should be excluded by histologic evaluation and immunotyping. Differential considerations also include metastases, which should be excluded by review of the medical history or im-

aging evidence of a separate primary neoplasm, typically accomplished with fluorodeoxyglucose (FDG) PET/CT. After these possible mimics are excluded, a diagnosis of iMCD can then be made on the basis of fulfillment of the major and minor criteria discussed (Table 2) (23).

Imaging Features

Diagnostic

CD most commonly manifests as unicentric disease with various sites of involvement. The chest is the most common site (30%–70%), followed by the neck (10%–40%), abdomen and pelvis (12%–39%), and axilla (4%–5%) (4,7,11,14,29). While lymph node involvement is typical, extralymphatic sites of possible disease include the lungs, pleura, pericardium, adrenal gland, larynx, retropharyngeal space, parotid glands, pancreas, meninges, muscles, and bones (Fig S1) (11,29,30).

CT is the preferred modality for evaluating patients with suspected CD and is often the first tool used in the diagnostic workup. Although the US appearance of CD is relatively nonspecific, the versatility, convenience, and availability of US enable rapid evaluation of palpable lesions in easily reached areas such as the neck, axilla, and groin. US can also be used for tissue sampling. MRI is a valuable anatomic tool in delineating the extent of masses and involvement of

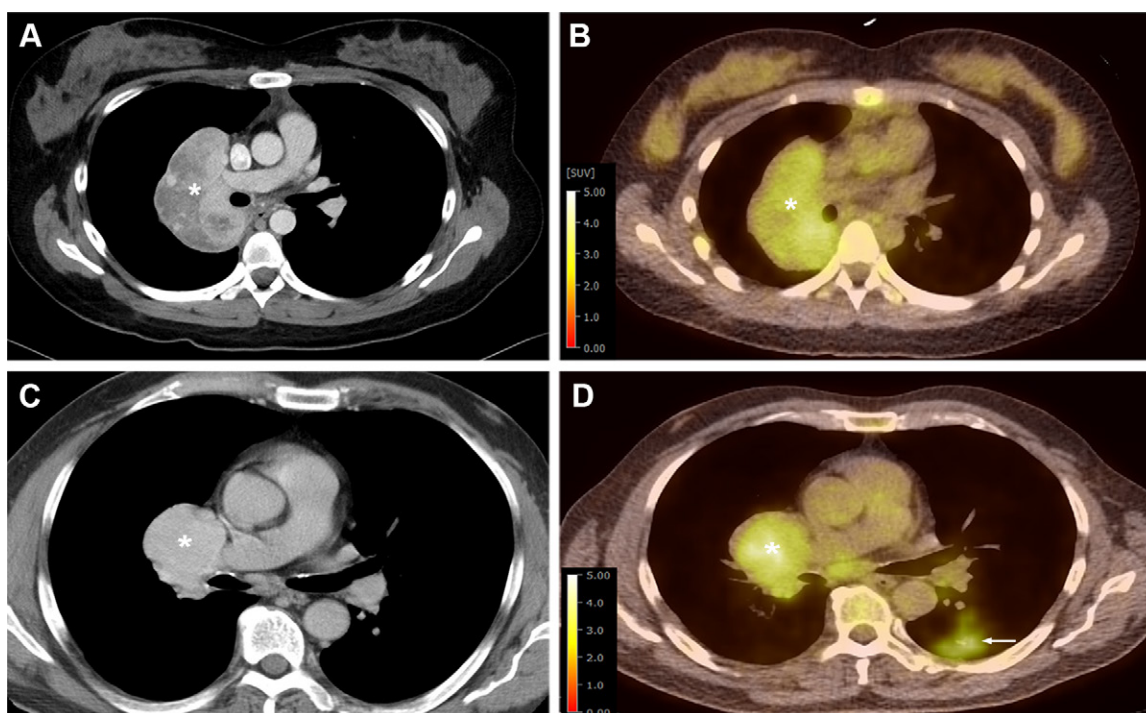


Figure 2. Two examples of thoracic UCD. (A, B) HVV CD in a 25-year-old woman who presented to the emergency department with chest pain. Axial contrast-enhanced CT image (A) shows a heterogeneously enhancing mass (* in A) centered at the right hilum with extension along the right mediastinum, which does not appear to invade adjacent structures. Subsequent axial FDG PET/CT image (B) shows that the mass (* in B) is FDG avid. Surgical excision revealed HVV CD. (C, D) HVV CD in a 65-year-old man who underwent CT after a right hilar mass was incidentally detected at screening chest radiography. Axial contrast-enhanced CT image (C) shows a homogeneously enhancing right hilar mass (* in C). Subsequent axial FDG PET/CT image (D) shows increased FDG activity in the mass (* in D). Surgical excision revealed HVV CD. Of note, the FDG activity identified in the posterior left lung (arrow in D) is due to infection or inflammation.

nearby structures (29,31). Conventional radiography is not very helpful in workup of CD, as findings are often nonspecific and noncontributory.

Unicentric CD

The most common manifestation of UCD is an incidental mass in the chest in an otherwise asymptomatic patient (11). Typical thoracic UCD occurs in the mediastinum or hilar region as a hypervascular round mass (29), which may be non-invasive in 50% of cases or invade adjacent structures in 40% (Fig 2). Hypervascular thoracic adenopathy in an asymptomatic patient without a known primary neoplasm is most suggestive of CD. Less commonly, thoracic UCD occurs as matted lymphadenopathy confined to one mediastinal compartment without a dominant mass (11,29).

Abdominal or pelvic UCD typically manifests as a solitary soft-tissue mass with well-defined margins (Fig 3) (28). These masses can occur in the mesentery, retroperitoneum, or both. The most frequent manifestation of abdominal UCD is an enlarged lymph node at the porta hepatis (32). Rare cases of gastrointestinal tract involvement, including the stomach, duodenum, and colon, as well as involvement of the pancreas have been reported (33–36).

Masses in UCD tend to be larger than those in MCD, with a mean diameter of 5.5 cm versus 2.3 cm in one study (30). This may be due to the relatively increased frequency of HVV CD in these patients with associated increased blood

supply. At noncontrast CT, the masses are usually homogeneous and appear hypo- or isoattenuating compared with skeletal muscle.

After administration of contrast material, most lesions show homogeneous enhancement, particularly when they are smaller than 5 cm and confined to the chest (29). Less commonly, lesions may show heterogeneous enhancement (Fig 2). The inconsistent enhancement pattern has been attributed to variable degrees of intralesion fibrosis, necrosis, and degeneration in lesions larger than 5 cm or those located in the abdomen or pelvis (29,30).

At US, involved lymph nodes or masses appear as well-defined homogeneously hypoechoic structures with posterior acoustic enhancement (Figs 4, 5) (14,29). The presence of associated vascularity at Doppler imaging is variable. When there is lack of vascularity, the lesions may mimic cysts (14).

The value of MRI in CD lies in its ability to provide multiplanar anatomic delineation of masses and their relationship with adjacent structures, as well as highlight the presence of feeding vessels (11,29). Affected lymph nodes or masses in CD are most frequently hypointense to isointense to skeletal muscle on T1-weighted images and isointense to hyperintense to skeletal muscle on T2-weighted images (Figs 6, 7) (11,14,29,30). As with CT, the enhancement pattern at MRI after intravenous administration of gadolinium contrast material can be homogeneous or heterogeneous (Figs 6, 7) (11,29,30). The enhancement can persist into the

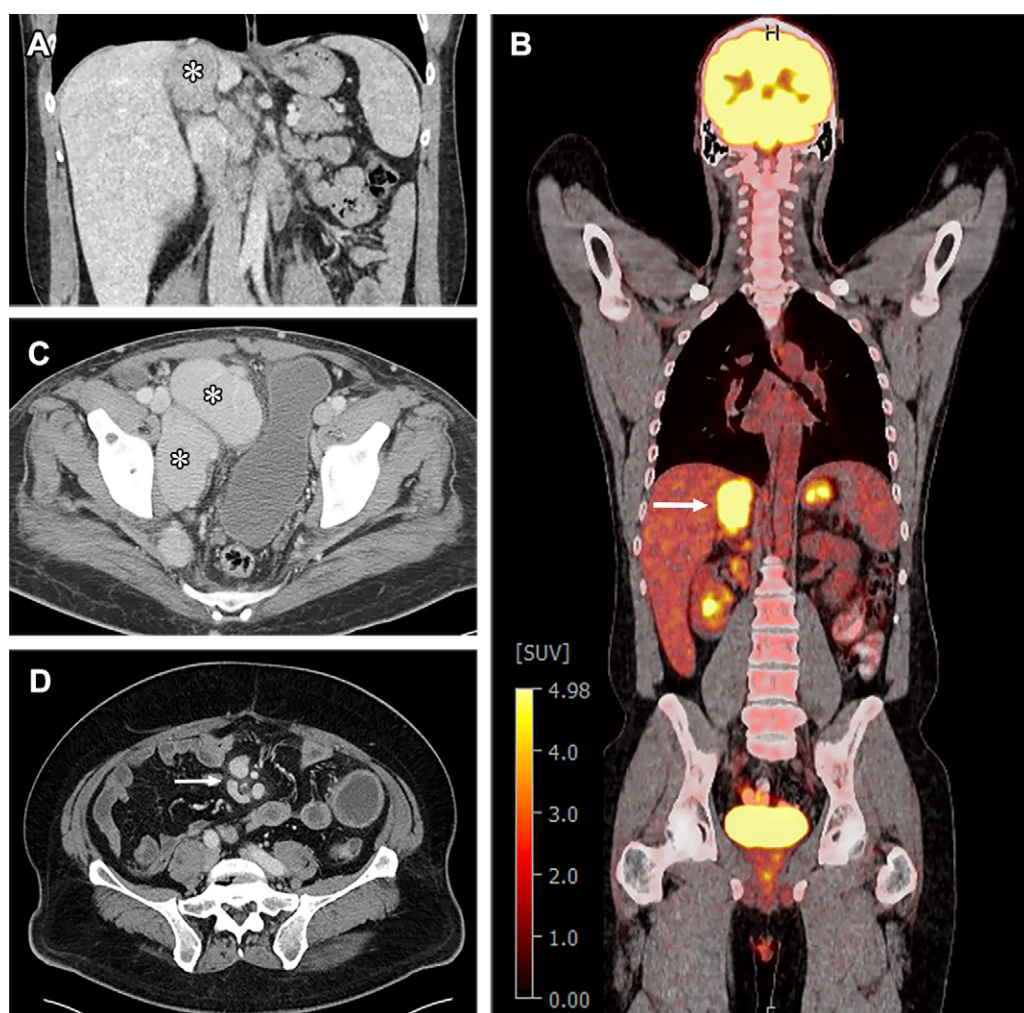


Figure 3. Manifestations of UCD in the abdomen and pelvis. (A, B) Mixed HVV and PCV CD in a 42-year-old man. Coronal contrast-enhanced CT image (A) shows a homogeneously enhancing enlarged portacaval lymph node (* in A). Subsequent coronal FDG PET/CT image (B) shows the node (arrow in B), which is FDG avid. Pathologic analysis revealed mixed HVV and PCV CD. The patient underwent combined US- and CT-guided ablation, with a decrease in the size of the mass at follow-up CT (Fig S9). (C) Mixed HVV and PCV CD in a 45-year-old asymptomatic man. Axial contrast-enhanced CT image shows homogeneously enhancing waxing and waning lymph nodes (*) confined to the right iliac chain. Pathologic analysis revealed mixed HVV and PCV CD. (D) HVV CD in a 50-year-old man. Axial contrast-enhanced CT image shows focally enlarged mesenteric lymph nodes (arrow). These were determined to be HVV CD after biopsy.

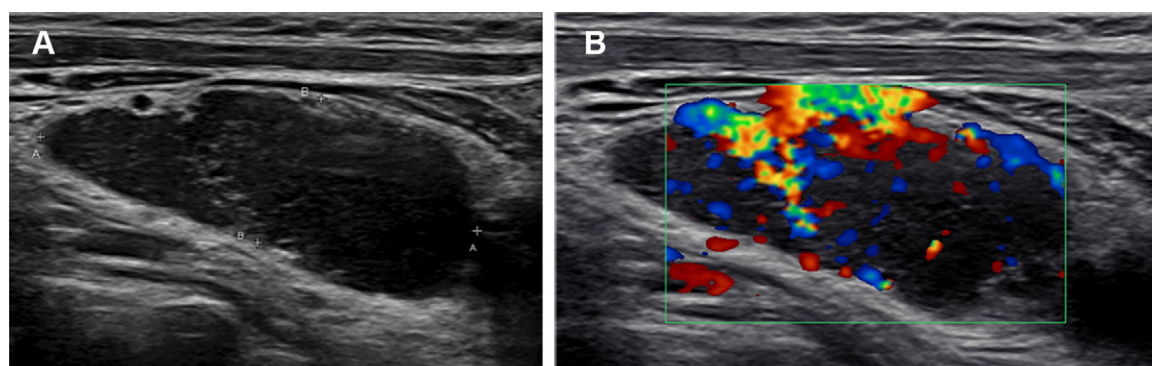


Figure 4. HVV CD in a 32-year-old man. (A) Gray-scale US image shows a 4.6-cm predominantly hypoechoic right supraclavicular lymph node. (B) On a color Doppler image, the node has avid internal vascularity. Excisional biopsy demonstrated HVV CD.

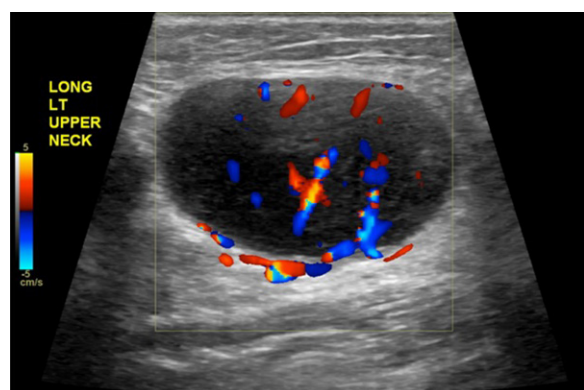


Figure 5. HVV CD in an 85-year-old woman with a history of EBV-positive B-cell lymphoproliferative disorder. Gray-scale US image of the left upper neck shows a hypoechoic 2.5-cm mass with internal vascularity. Biopsy results were compatible with HVV CD.

portal venous phase. Masses are typically hyperintense at diffusion-weighted imaging (30,36).

Multiple modalities may demonstrate prominent feeding vessels to the lymph node or mass, which can affect the surgical approach (29). At noncontrast MRI, these feeding vessels may appear as flow voids on T2-weighted images (11,29). The presence of a feeding vessel is sometimes considered a relatively specific feature of UCD, although this association is variable (28).

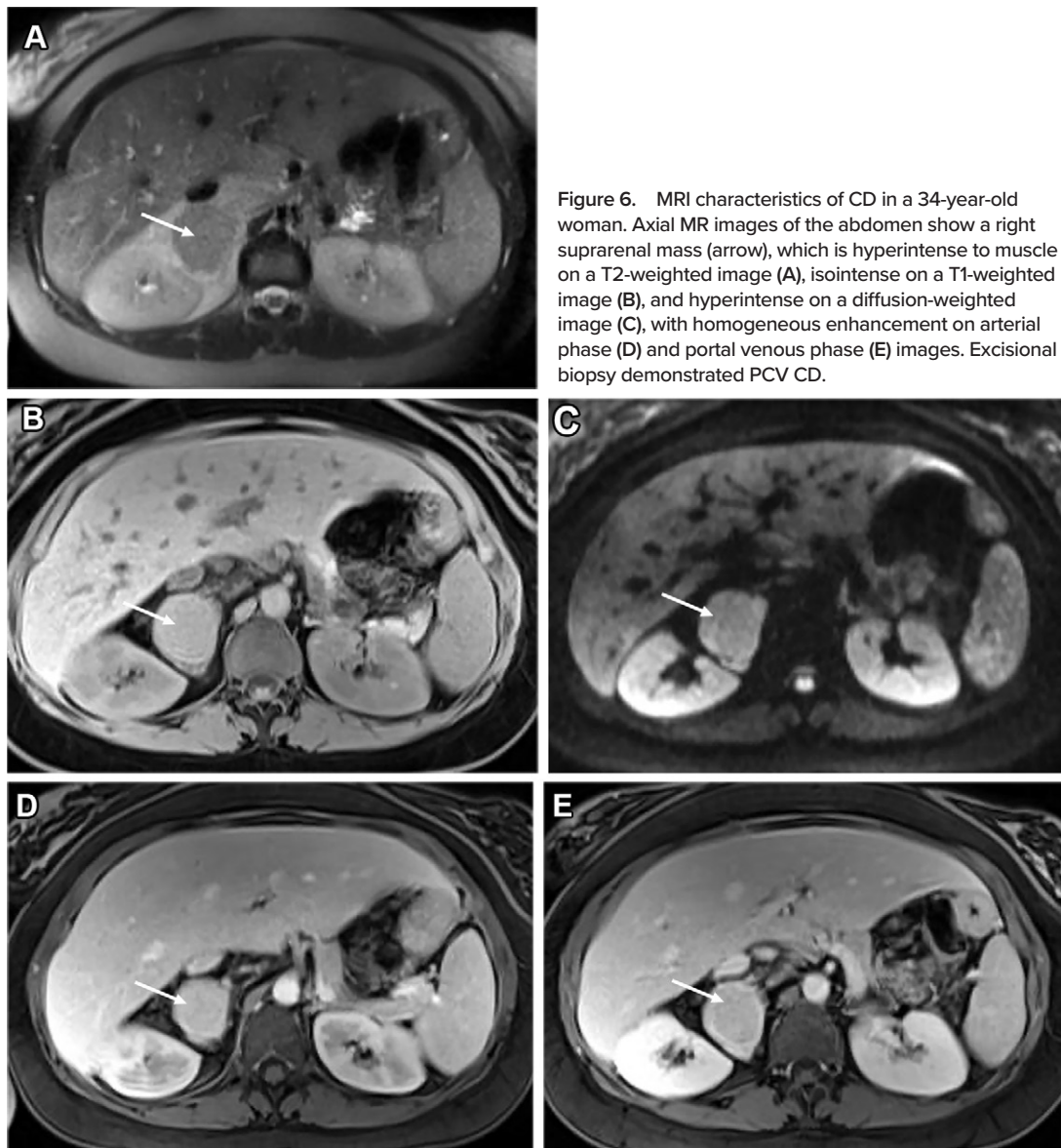


Figure 6. MRI characteristics of CD in a 34-year-old woman. Axial MR images of the abdomen show a right suprarenal mass (arrow), which is hyperintense to muscle on a T2-weighted image (A), isointense on a T1-weighted image (B), and hyperintense on a diffusion-weighted image (C), with homogeneous enhancement on arterial phase (D) and portal venous phase (E) images. Excisional biopsy demonstrated PCV CD.

Calcifications in CD are relatively rare, occurring in approximately 10% of cases (11). While it has been suggested that the presence of calcification may indicate a diagnosis of UCD, this has not been consistently proven (14,28,30). Calcifications in CD can be discrete, be coarse, or have an arborizing configuration, but they are usually nonspecific (Figs 8, S2) (11). At US, associated calcifications may appear as echogenic shadowing foci (Fig 8) (28). Visualization or characterization of calcifications at MRI is limited (29).

Multicentric CD

Lymph nodes in MCD tend to be hypervascular, similar to those in UCD, although some reports suggest that nodal masses associated with HHV8-MCD may not show substantial enhancement (10). MCD in the chest typically manifests as cervical, hilar, mediastinal, or axillary lymphadenopathy, which affects multiple nodal chains (Figs 9, 10) (11,29). Pleural or pericardial effusions are common (Figs 11, 12). Lung involvement may appear as centrilobular nodules and

nodular opacities and less frequently as ground-glass attenuation, airspace consolidation, and bronchiectasis (Fig 13) (29). Although unusual, when direct pleural involvement occurs, it can manifest as a well-defined pleural-based mass with or without associated pleural effusion (Fig S3) (11).

In the abdomen, MCD commonly manifests as diffuse lymphadenopathy (Fig 9), but it can also demonstrate hepatosplenomegaly, ascites, and anasarca (Figs 11, 12); thickening of the retroperitoneal fascia can infrequently occur (29). Hepatosplenomegaly typically has homogeneous parenchymal signal intensity on T2-weighted, T1-weighted, and diffusion-weighted images (32).

Osseous lesions are common in MCD patients with POEMS syndrome (Fig 14). The lesions are most commonly sclerotic but may also be lytic or mixed in appearance. Scrutiny of bone at CT in patients with MCD is thus important, as the presence of these lesions may suggest the diagnosis of POEMS-MCD.

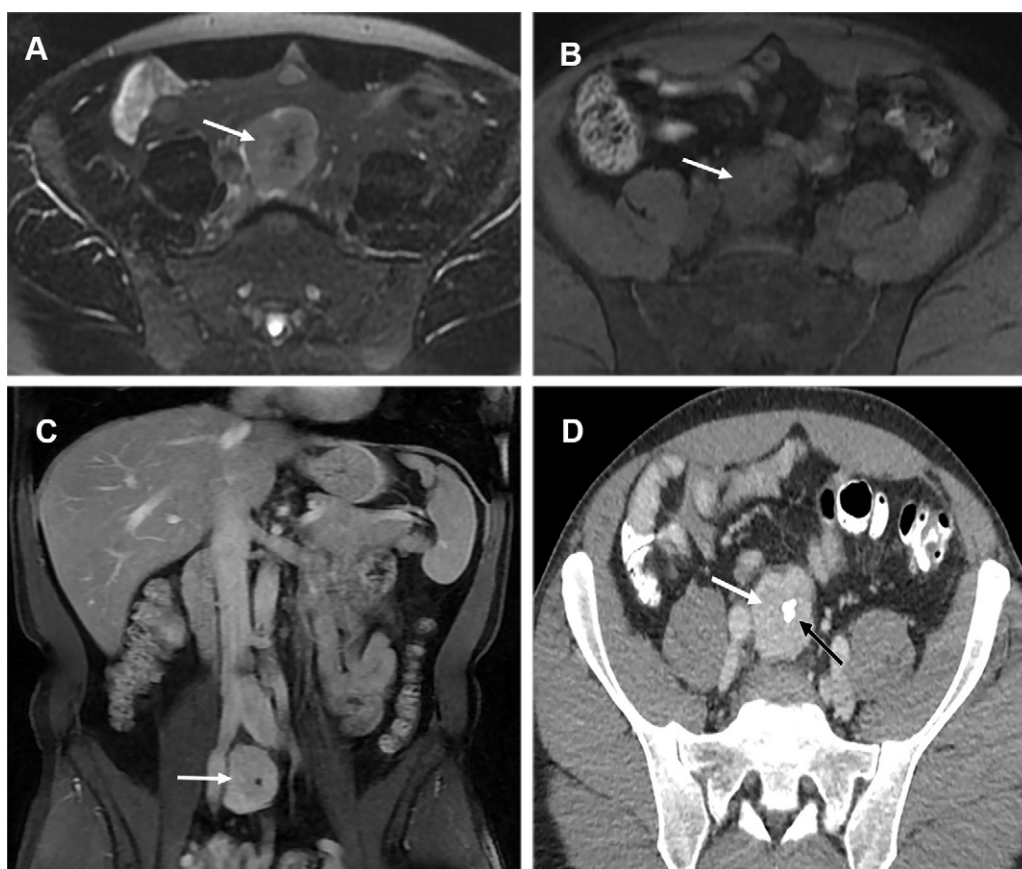


Figure 7. MRI characteristics of CD in a 46-year-old man. (A–C) MR images of the pelvis show a homogeneous mass (arrow) at the aortic bifurcation, which is moderately hyperintense on an axial T2-weighted image (A), hypointense on an axial T1-weighted image (B), and avidly enhancing on a coronal contrast-enhanced image (C). A central hypointensity is present with all sequences. (D) Axial contrast-enhanced CT image shows that the central hypointensity in A–C corresponds to a coarse central calcification (black arrow) in the mass (white arrow). Biopsy demonstrated HVV CD. The patient underwent surgical resection, with no residual disease at subsequent PET/CT (not shown).

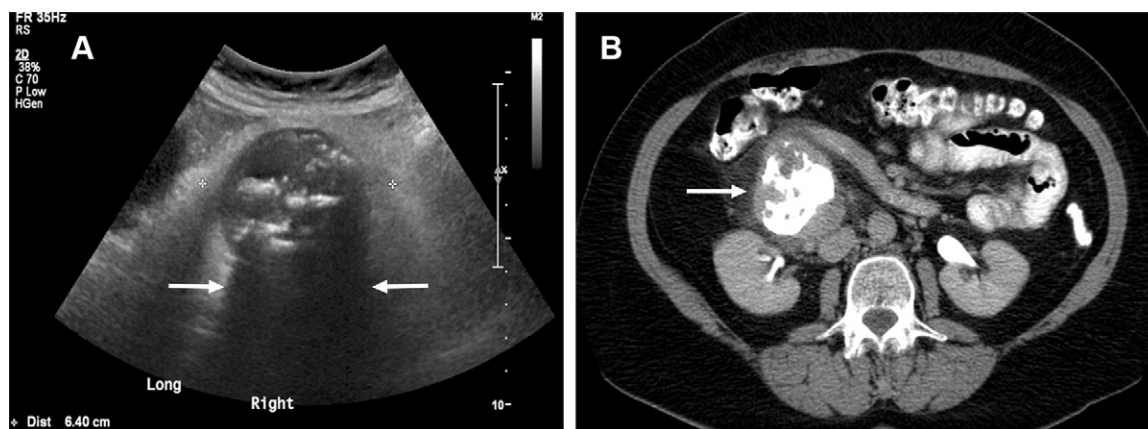


Figure 8. HVV CD in a 32-year-old asymptomatic man who presented for an annual physical examination. (A) Gray-scale US image of the right upper abdomen shows a hypoechoic mass with scattered internal calcifications, which demonstrate posterior acoustic shadowing (arrows). (B) Subsequent axial contrast-enhanced CT image shows a corresponding calcified right pararenal mass (arrow), with calcifications arising in an arborizing configuration. Pathologic analysis demonstrated HVV CD.

Positron Emission Tomography

PET/CT is an incredibly useful tool for patients with known or suspected CD. In particular, FDG PET/CT can assist in selecting the most suitable lymph node for biopsy on the basis of size and maximum standardized uptake value (SUV_{max}), demonstrate the extent of disease, and allow monitoring of patients with confirmed CD (31,37). Typically, involved lymph nodes or masses are mild to moderately hypermetabolic at PET/CT (Figs 2, 3, 10, 12, 14), with a mean SUV_{max} between 5.6 and 5.8 (range, 2–19) (38,39). Increased met-

abolic activity in the spleen may also be observed at FDG PET/CT (38).

It has been suggested that SUV_{max} may be higher in MCD versus UCD and in patients with active clinical symptoms (39). However, in comparison with that of any potential disease mimics, the median SUV_{max} of lymph nodes involved by CD tends to be slightly lower. The presence of more active lesions may indicate an alternative diagnosis, such as high-grade lymphoma (31). In the setting of POEMS syndrome, osseous lesions may also be FDG avid (40).

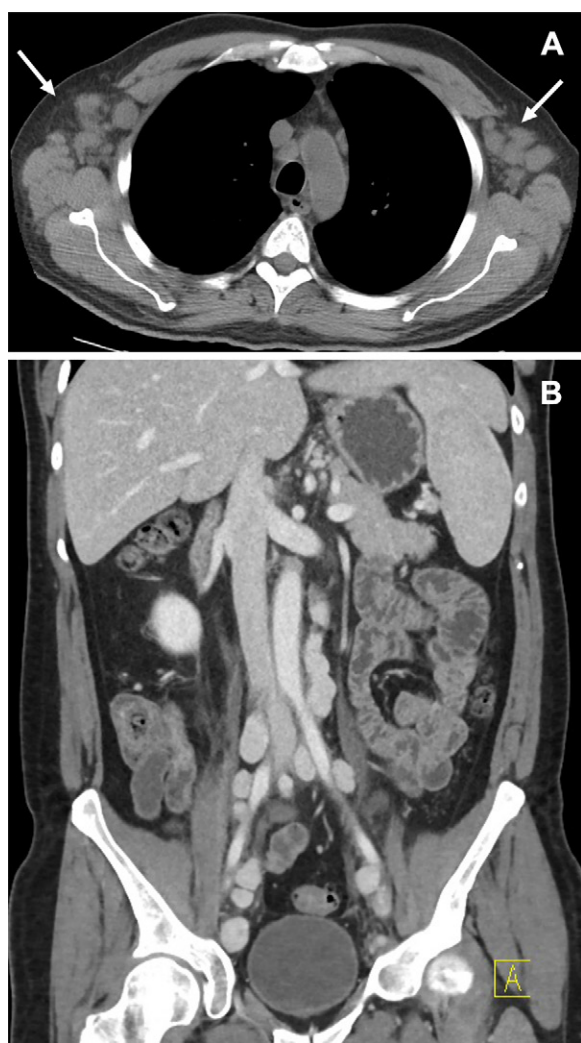


Figure 9. Lymphadenopathy in the setting of HHV8-MCD in a 65-year-old man who was positive for HIV. (A) Axial noncontrast CT image of the chest shows axillary lymphadenopathy (arrows). (B) Coronal contrast-enhanced CT image of the abdomen and pelvis shows retroperitoneal lymphadenopathy. The patient was positive for HHV8; pathologic analysis demonstrated PCV CD. The findings were compatible with HHV8-MCD.

Mimics and Differential Diagnosis

The imaging features of CD may overlap with those of other conditions, making it difficult to arrive at a definitive diagnosis on the basis of imaging features alone. CD should be considered in the differential diagnosis of patients presenting with nodal and extranodal masses, generalized lymphadenopathy, and hyperenhancing lymph nodes, in addition to those with a working diagnosis of lymphoma or systemic infection (11). CD should be a particular consideration in patients who are immunocompromised (11). Therefore, biopsy is typically performed before treatment planning to confirm the diagnosis.

Lymphoma is a common mimic that is often a diagnostic consideration for presumed CD in different parts of the body (Fig 15). Lymphoma can be confused with CD in the mediastinum, MCD, or nonspecific masses in other parts of the body (11). At imaging, increasing adenopathy, ascites, or new splenomegaly may indicate a diagnosis of lymphoma in the setting of known CD (14). Nodal masses with higher SUV_{max} are also

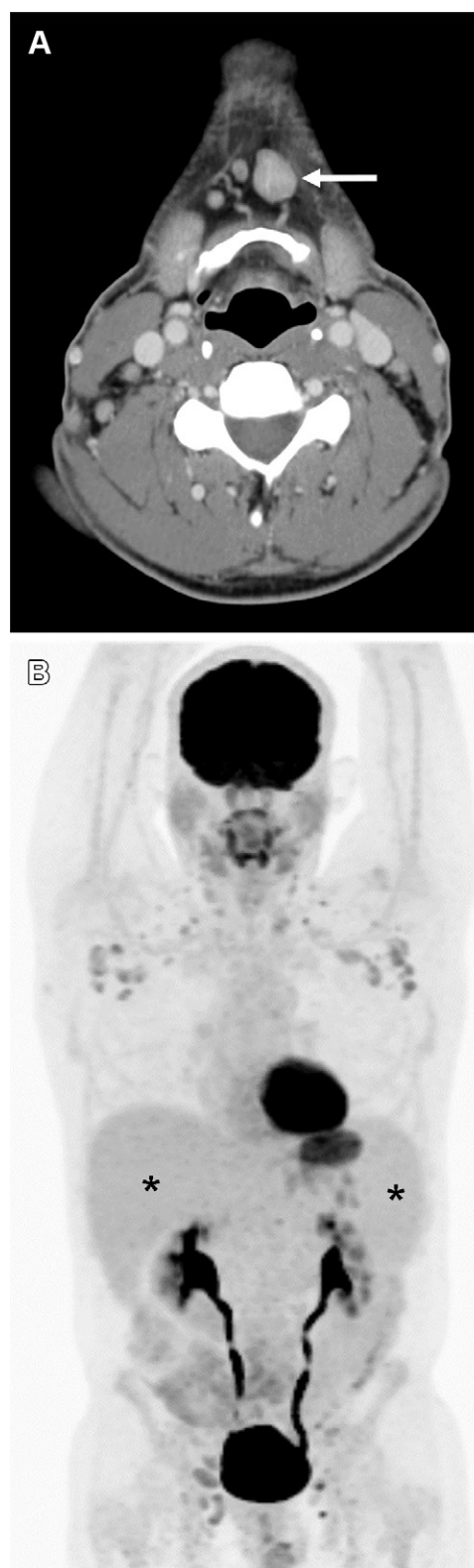


Figure 10. PCV CD in a 38-year-old man positive for HIV with a history of Kaposi sarcoma who presented with fever, night sweats, and fatigue. (A) Axial contrast-enhanced CT image of the neck shows enlarged cervical lymph nodes (arrow). (B) Subsequent coronal FDG PET/CT image shows widespread cervical, supra- and infraclavicular, axillary, and inguinal lymphadenopathy, in addition to hepatosplenomegaly (*). Pathologic analysis revealed PCV CD.

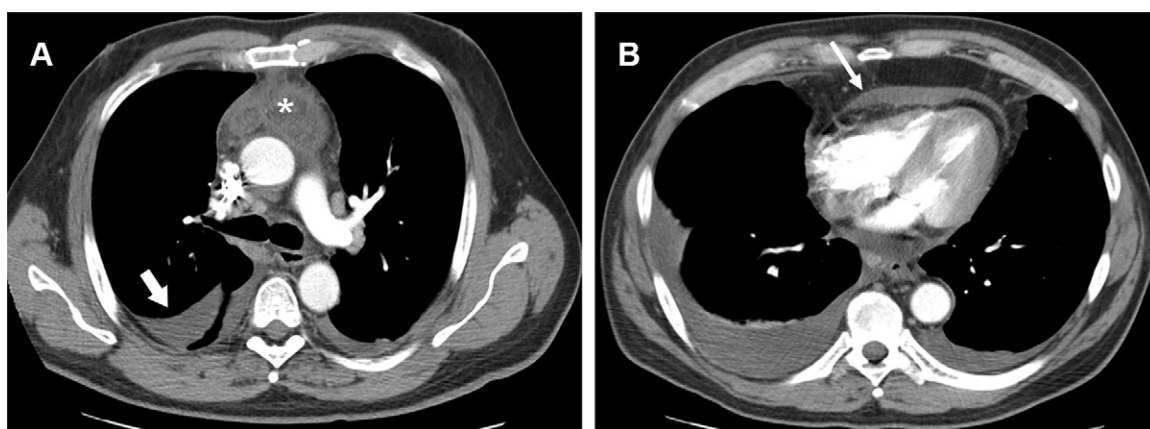


Figure 11. HVV CD in a 76-year-old man. Axial contrast-enhanced CT images of the chest show a mediastinal mass (* in A) of heterogeneous attenuation and diffuse anasarca, characterized by large right-greater-than-left pleural effusions (arrow in A), ascites (not shown), and a pericardial effusion (arrow in B). Biopsy revealed HVV CD.

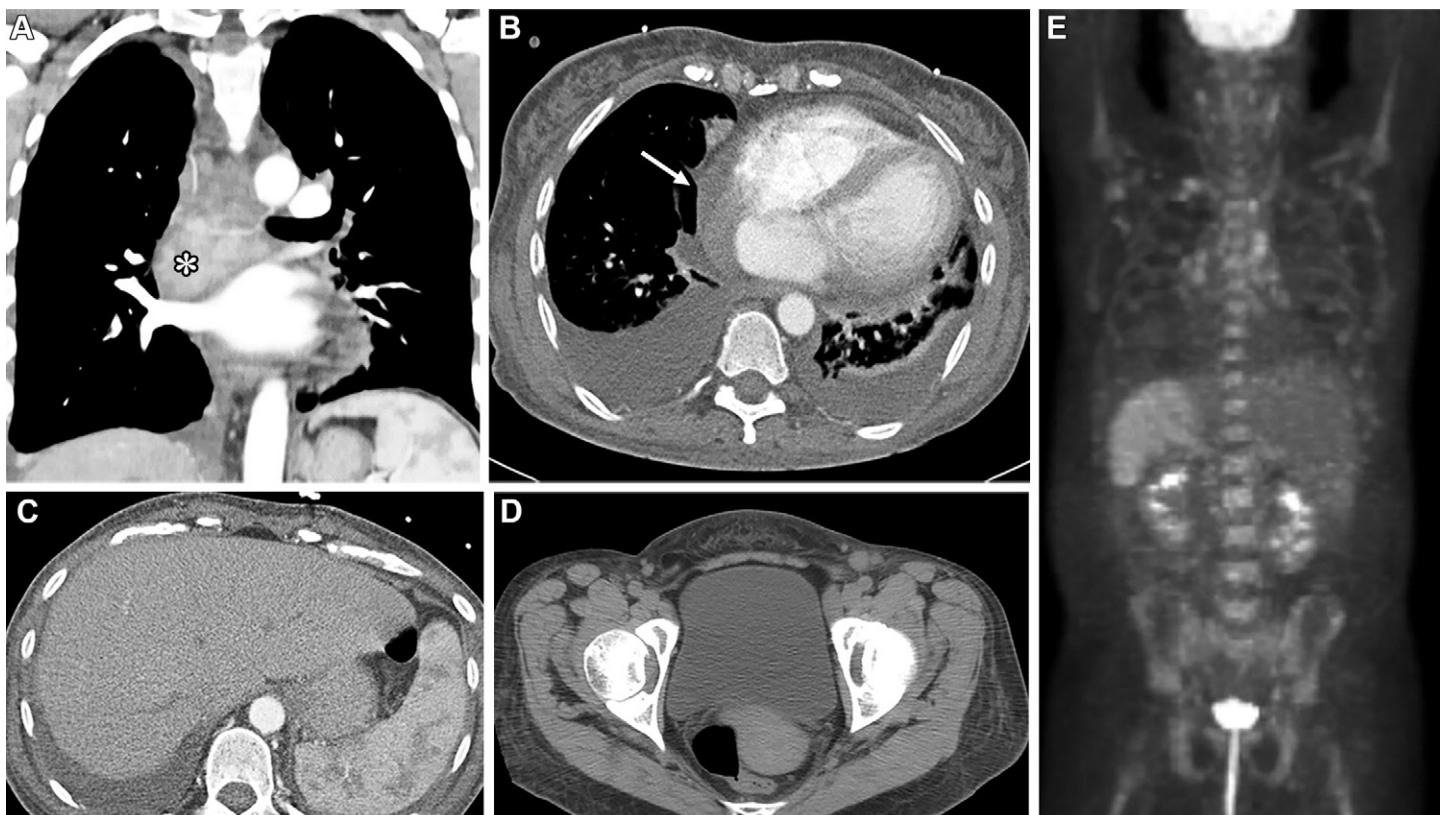


Figure 12. PCV CD in a 68-year-old woman. (A, B) Coronal (A) and axial (B) contrast-enhanced CT images of the chest show a mediastinal mass (* in A) as well as large bilateral pleural effusions and a pericardial effusion (arrow in B). The patient also had axillary lymphadenopathy (not shown). (C) Axial contrast-enhanced CT image of the abdomen shows hepatosplenomegaly. (D) Contrast-enhanced CT image of the pelvis shows inguinal lymphadenopathy and generalized anasarca with diffuse body wall edema. (E) Coronal FDG PET/CT image shows diffuse marrow uptake, particularly within the vertebral column. Biopsy of the mediastinal mass demonstrated PCV CD.

concerning for either non-Hodgkin lymphoma (NHL) or Hodgkin lymphoma, as the median SUV of lymph nodes involved by CD tends to be slightly lower.

The presence of calcification within a lesion would argue against a diagnosis of lymphoma, since calcification is rare in cases of untreated lymphoma (14). Lymphoma can also develop subsequently in patients with CD (Figs 16, 17). Overall, given the significant overlap between the clinical manifestations and imaging features of lymphoma and CD, diagnosis requires a tissue biopsy (14).

As discussed earlier in the section on diagnostic criteria for possible iMCD, potential mimics for diffuse or multisystem dis-

ease include infectious causes such as infectious mononucleosis, cytomegalovirus (CMV), and HIV as well as autoimmune causes such as rheumatoid arthritis, systemic lupus erythematosus (SLE), adult-onset Still disease, and juvenile idiopathic arthritis (JIA). Sarcoidosis can mimic several manifestations of CD (Fig S4). These diagnoses are largely differentiated from CD on the basis of clinical and laboratory findings (Table 3).

Other potential mimics of CD include multiple myeloma and solitary plasmacytoma (Fig 18), including both the extramedullary and intraosseous subtypes (Table 3) (23).

Like UCD, FDC sarcoma manifests as a well-defined and hypervascular solitary mass (41). When confined to lymph nodes,

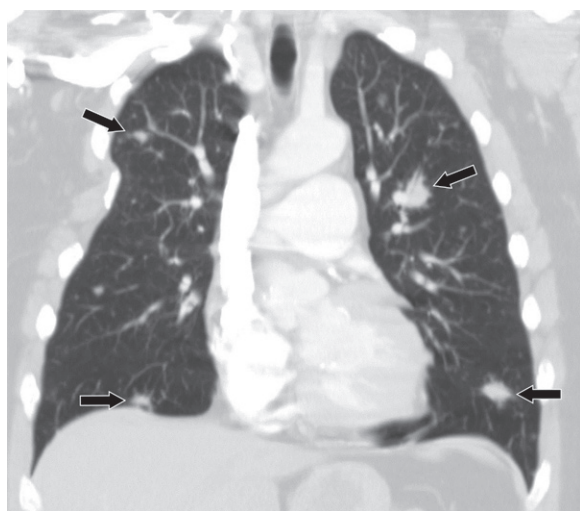


Figure 13. Lung involvement by CD in a 78-year-old woman. Coronal noncontrast CT image of the chest shows multifocal lung nodules (arrows). Biopsy revealed PCV CD.

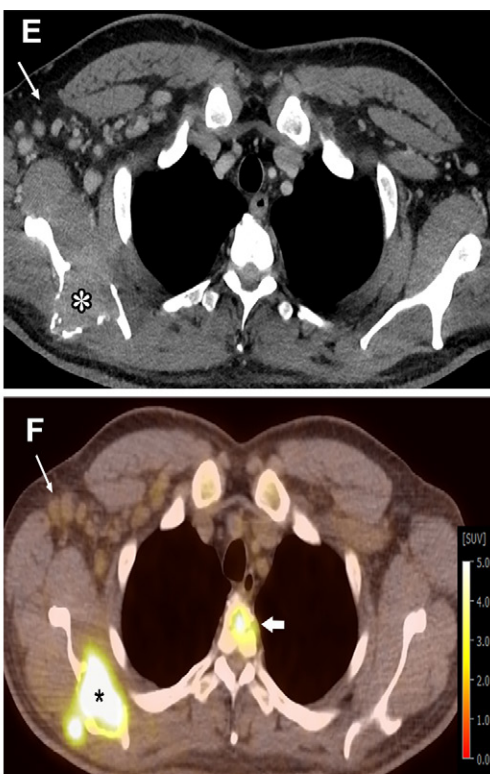
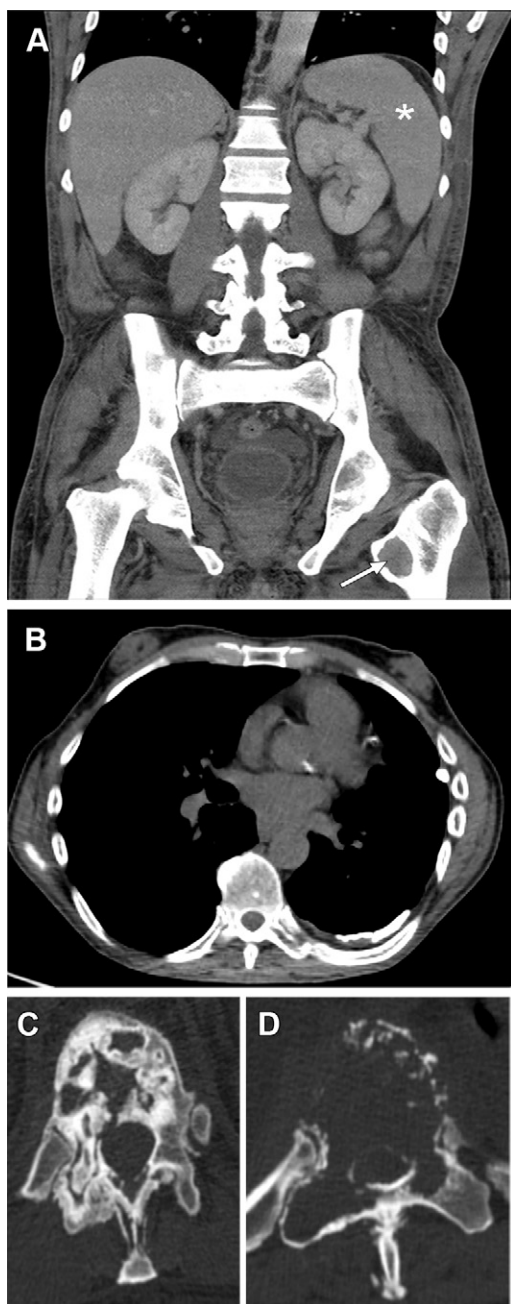


Figure 14. (A, B) Manifestations of POEMS-MCD in a 39-year-old man. Coronal CT image of the chest, abdomen, and pelvis (A) and axial CT image of the chest (B) show splenomegaly (* in A), a lytic lesion in the left lesser trochanter (arrow in A), and hypogonadism related to endocrinopathy from POEMS syndrome manifesting as bilateral gynecomastia (B). Nonpictured manifestations of POEMS syndrome in this patient included sensorimotor neuropathy, lymphadenopathy, cherry angiomas, and bilateral deep venous thrombosis. (C, D) Osseous lesions in a 35-year-old man with POEMS-MCD. Axial CT images of the spine show a mixed lytic and sclerotic lesion (C) and a lytic lesion (D) in two different thoracic vertebral bodies. The diagnosis of CD was based on results of lymph node biopsy. (E, F) Manifestations of POEMS-MCD in a 46-year-old man. (E) Axial CT image of the chest shows a lytic lesion (*) in the right scapular body. Arrow = right axillary lymphadenopathy. (F) On a subsequent axial FDG PET/CT image, the lytic lesion (*) has increased activity. Note the additional FDG-avid osseous lesion in the spine (thick arrow), which was not evident at prior noncontrast CT, as well as the right axillary lymphadenopathy (thin arrow). Lymph node biopsy demonstrated CD.

FDC sarcoma tends to be smaller and more homogeneous. When extranodal, FDC sarcoma is usually larger and more heterogeneous with internal necrosis and regional adenopathy (41). FDC sarcoma can also have coarse arborizing calcifications and feeding vessels. The main distinction between FDC sarcoma and CD is that masses in FDC sarcoma can show progressive enhancement in delayed phases, which has not been observed in CD.

At imaging, other soft-tissue sarcomas—including undifferentiated pleomorphic sarcoma, solitary fibrous tumor, and gastrointestinal stromal tumor (GIST)—may also resemble retroperitoneal or mesenteric CD (28). In the neck and mediastinum, imaging findings of CD can be mistaken for those of thymic epithelial neoplasms (Fig S5), germ cell tumors, and neural crest-derived neoplasms such as paraganglioma (Fig 19), neurofibroma, or schwannoma (11).

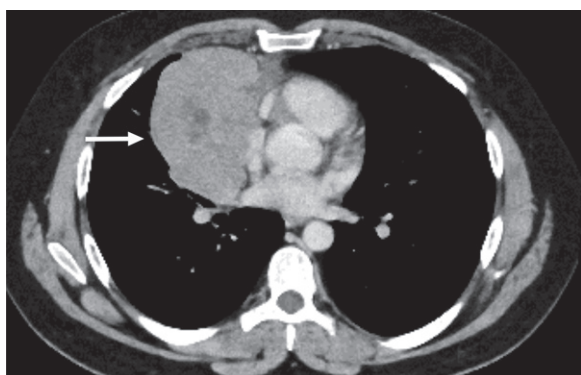


Figure 15. Lymphoma as a mimic of CD in a 48-year-old man who presented with repeated upper respiratory tract infection, persistent cough, and recent development of night sweats. Axial contrast-enhanced CT image shows a large lobulated mass (arrow) in the right hemithorax abutting the pericardium without any signs of local invasion. Note the heterogeneous attenuation of the mass with central hypoattenuation. Biopsy demonstrated large B-cell lymphoma.

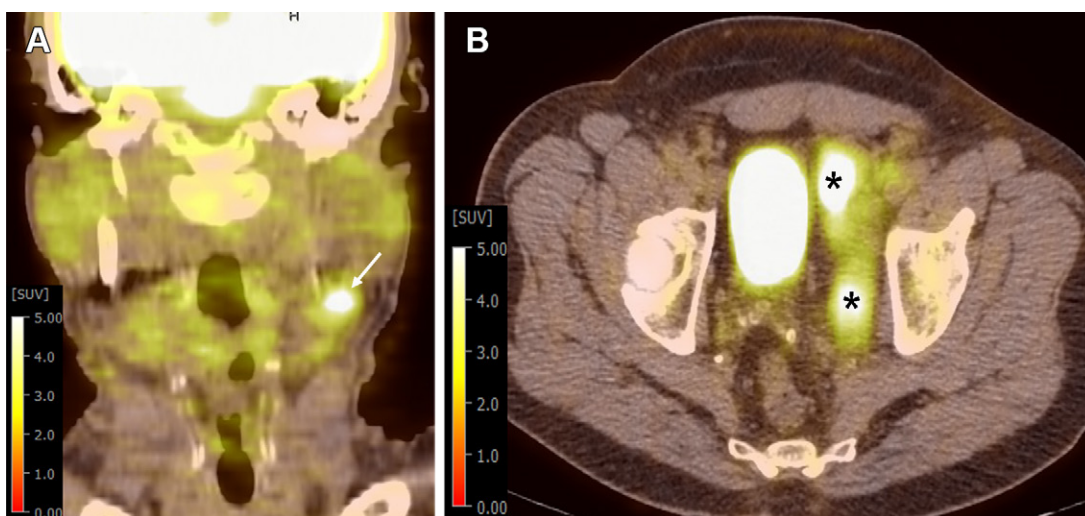


Figure 16. PCV CD with subsequent development of lymphoma in a 70-year-old man who initially presented with an enlarged cervical lymph node. (A) Coronal FDG PET/CT image shows an enlarged and FDG-avid left cervical lymph node (arrow). (B) Follow-up axial FDG-PET/CT image shows enlarged and FDG-avid left iliac lymph nodes (*). Excisional biopsy of a left iliac lymph node demonstrated PCV CD with additional CD5-positive marginal zone B-cell proliferation, compatible with marginal zone lymphoma.

Like with CD in the chest, neurogenic and nerve sheath tumors (Figs 20, 21), paraganglioma (Fig 22), and pheochromocytoma (Fig S6) are also diagnostic considerations when evaluating UCD in the abdomen or pelvis (11). Lesions located near the pancreas may be mistaken for pancreatic neuroendocrine tumors at CT or MRI (11,33). An exophytic liver mass (Fig S7) and a hyperenhancing pancreatic mass (Fig S8) can mimic CD. The clinical and imaging differences between these diagnostic possibilities are highlighted in Table 3.

Treatment and Posttreatment Response

Management of UCD is typically surgical. Complete surgical excision is usually curative with an excellent prognosis (1,4). When surgery is not feasible owing to patient or location-related factors, alternative treatment options include radiation therapy, embolization, or neoadjuvant therapy with rituximab or with siltuximab or tocilizumab (Fig S9) (1).

MCD typically requires systemic therapies; in general, no treatment differences exist between MCD pathologic subtypes. Siltuximab, tocilizumab, and/or rituximab—with or without steroids—are often first-line approaches. When the patient is critically ill, for example with TAFRO, systemic chemotherapy is often used. Although there is no consensus on the best chemotherapy approach, lymphoma-type or myeloma-type regimens are often employed (1).

Responses to anti-interleukin 6 (IL-6) therapy may be delayed compared with responses to chemotherapy, and overall

response to treatment is usually assessed using clinical symptoms and laboratory values such as hemoglobin, erythrocyte sedimentation rate (ESR), C-reactive protein (CRP) level, and albumin level (1). HHV8-MCD often responds well to rituximab-based therapy. Antiretroviral therapy is essential when HHV8-MCD is driven by coexistent HIV infection. Patients with POEMS-MCD presenting with osseous involvement can be treated with standard myeloma therapy, preferably high-dose chemotherapy with autologous stem cell transplant (ASCT), and may require plasma cell-directed systemic therapy (Fig S10) (1).

Prognosis

In the absence of complications or development of lymphoma, UCD typically has no significant effect on life expectancy (5). Available data on outcomes and prognosis in MCD are more limited. The clinical course of patients with MCD is variable: some patients may present with indolent and slowly progressive disease, others may experience a relapsing-remitting course, and still others may present with fulminant disease that can be fatal within the course of weeks (7). In one study, 35% of patients diagnosed with iMCD were reported to have died within 5 years of diagnosis and 60% were reported to have died within 10 years. Patients also have a threefold increased prevalence of malignancy (23).

In another study, however, estimated 5-year overall survival among patients with iMCD was 100% (of note, however,

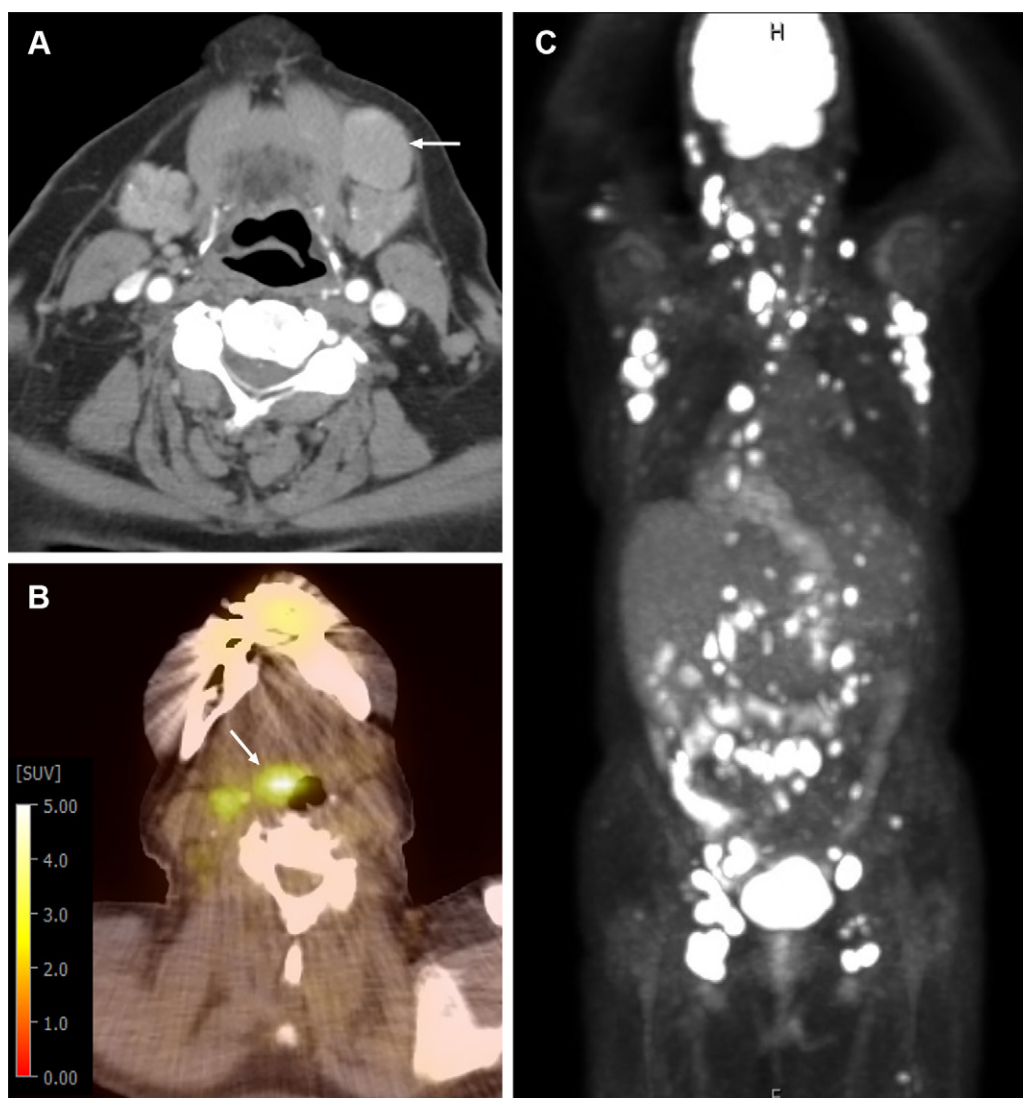


Figure 17. CD with subsequent development of lymphoma in an 85-year-old woman. (A) Axial contrast-enhanced CT image shows an enlarged submandibular lymph node (arrow). Excisional biopsy revealed HVV CD. (B) Axial FDG PET/CT image 1 year later shows an FDG-avid lesion (arrow) along the right tongue base. Biopsy demonstrated EBV-associated lymphoma. (C) Follow-up coronal FDG PET/CT image 1.5 years later shows diffuse lymphadenopathy related to lymphoma progression.

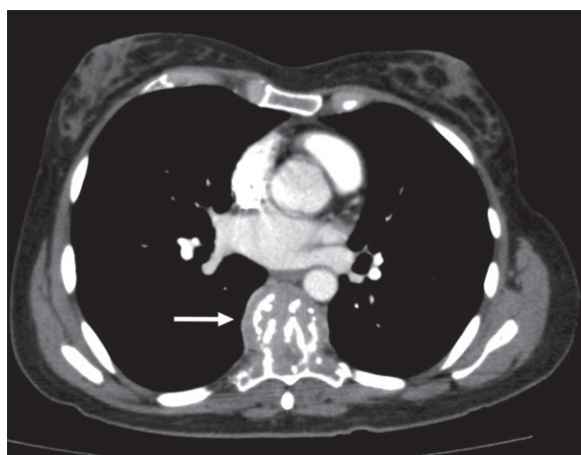


Figure 18. Plasmacytoma as a mimic of CD in a 36-year-old woman with back pain. Axial CT image of the chest shows a destructive lesion of the T8 vertebra (arrow) with associated paravertebral soft-tissue thickening, findings that could mimic the mixed sclerotic-lytic lesions of POEMS-MCD. Biopsy demonstrated plasmacytoma. Four years later, the patient was diagnosed with multiple myeloma.

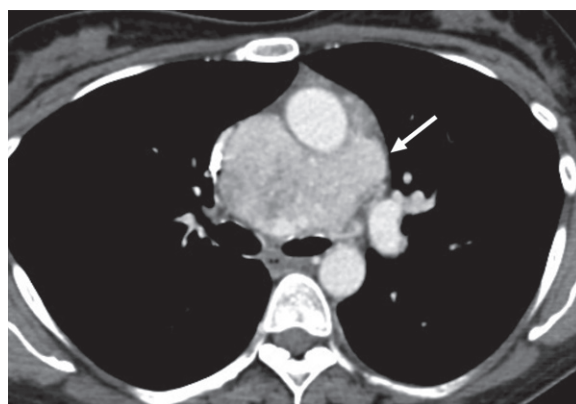


Figure 19. Paraganglioma as a mimic of CD in a 44-year-old woman with a family history of paraganglioma who presented with persistent cough. Axial contrast-enhanced CT image shows a large hyperenhancing mass (arrow) in the middle mediastinum that mildly compresses the tracheal bifurcation. Surgical excision demonstrated paraganglioma.

Table 3: Mimics of CD

Disease	Mimics What Sub-type of CD	Differentiating or Suggestive Clinical Features	Differentiating or Suggestive Imaging Findings
Lymphoma	UCD, MCD	Development of B symptoms or abnormal results of laboratory tests (eg, anemia)	Increasing adenopathy, ascites, or new splenomegaly Higher SUV_{max} than in CD Lack of calcification
Systemic infections (CMV-, HIV-, or EBV-associated disorders)	MCD	History of exposure Positive results of serologic tests Abnormal results of laboratory tests (eg, elevated WBC count)	...
Autoimmune disease (RA, SLE, adult-onset Still disease, JIA)	MCD	Positive results of serologic tests for disease (eg, ANA, RF)	...
Multiple myeloma (MM)	MCD (POEMS-MCD)	Bone pain in MM High incidence of renal failure in MM Better survival in CD	More likely lytic lesions in MM vs sclerotic lesions in CD
Solitary plasmacytoma (EMP, SBP)	UCD (EMP), MCD (POEMS-MCD, SBP)	EMP most frequently involves head and neck region, gastrointestinal tract, and lungs	SBP typically manifests as lytic osseous lesion
FDC sarcoma	UCD, MCD	Most FDC sarcomas occur in head and neck region; lesions arising in CD occur in abdomen	FDC sarcoma shows progressive enhancement in delayed phases
Thymic epithelial tumors (eg, thymoma)	UCD	Possible clinical evidence of MG	Hypoenhancing anterior mediastinal mass that may have cystic spaces or calcifications Invasion of great vessels, fat, or pleural seeding suggests malignancy
Germ cell tumor (eg, teratoma)	UCD	Younger patients (2nd to 4th decades of life)	Fluid, fat, $\pm$ calcification at CT and MRI No hyperenhancement
Soft-tissue sarcoma (eg, undifferentiated pleomorphic sarcoma)	UCD	Sarcomas tend to occur in younger patients	Origin from bone or muscle More invasive and locally destructive Iso- or hypoattenuating to adjacent muscle at CT
Solitary fibrous tumor (SFT)	UCD	...	SFT shows intermediate T1-weighted and heterogeneous T2-weighted signal intensity at MRI SFT may show avid enhancement
Gastrointestinal stromal tumor (GIST)	...	May manifest with bowel symptoms	Submucosal location Lymphadenopathy is rare
Neural crest-derived neoplasms (paraganglioma, pheochromocytoma)	UCD	Symptoms of catecholamine excess Association with NF types 1 and 2 Avidity at gallium 68-DOTATATE imaging or iodine 123-MIBG scintigraphy	Characteristic locations within paraganglia Cystic changes, necrosis, hemorrhage, and internal calcifications Salt-and-pepper enhancement due to prominent flow voids
Neurofibroma and schwannoma	...	Associated with NF types 1 and 2 Typical nerve distribution location	Fusiform shape, peripheral T2 hyperintensity with central hypointensity (target sign), and scalloping of adjacent vertebrae
Pancreatic neuroendocrine tumor	UCD	Symptoms related to hormonal excess if secreting hormones	Located in pancreas Avidity at somatostatin receptor imaging
Metastasis	UCD, MCD	Evidence of primary neoplasm elsewhere	Metastatic lymph nodes are usually hypoenhancing Presence of metastases in liver and lung
Sarcoidosis	MCD	Symmetric hilar and mediastinal lymphadenopathy Usually no mass effect on adjacent structures	Hypoenhancing nodes $\pm$ nodal calcification FDG avid and high uptake at gallium 67 scintigraphy Perilymphatic micronodules Hepatosplenomegaly with hypoenhancing granulomas Pulmonary fibrosis in late stages

Sources.—References 41–49.

Note.—ANA = antinuclear antibodies, CMV = cytomegalovirus, EMP = extramedullary plasmacytoma, JIA = juvenile idiopathic arthritis, MG = myasthenia gravis, MIBG = metaiodobenzylguanidine, NF = neurofibromatosis, $\pm$ = with or without, RA = rheumatoid arthritis, RF = rheumatoid factor, SBP = solitary bone plasmacytoma, SLE = systemic lupus erythematosus, SP = solitary plasmacytoma, SUV_{max} = maximum standardized uptake value, WBC = white blood cell.

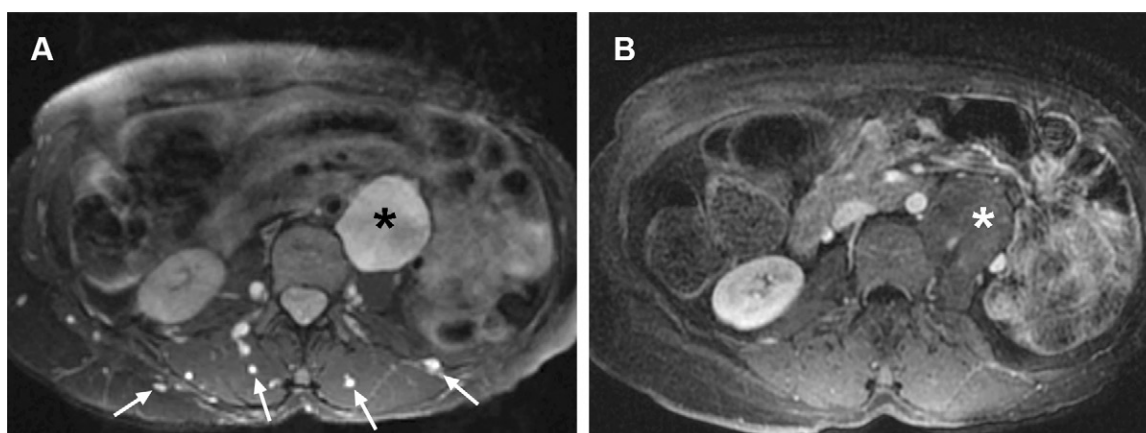


Figure 20. Neurofibroma as a mimic of CD in a 40-year-old woman with known neurofibromatosis. Axial MR images of the abdomen show a large neurofibroma (*) in the left para-aortic region, with hyperintensity on a T2-weighted image (A) and mild heterogeneous enhancement on a postcontrast T1-weighted image (B). Excision demonstrated a neurofibroma. Note the multiple small additional neurofibromas (arrows in A) in the back and subcutaneous fat, best seen on the T2-weighted image.

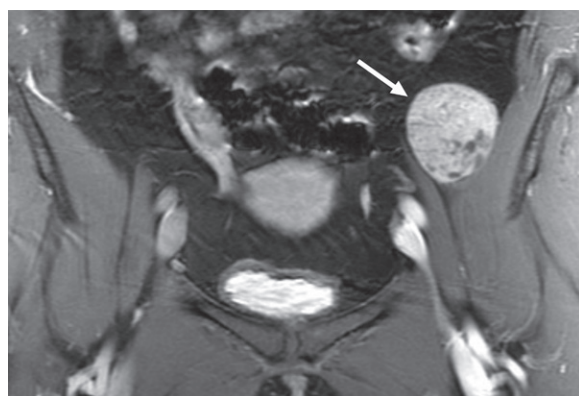


Figure 21. Schwannoma as a mimic of CD in a 45-year-old woman with occasional tingling and numbness in the left leg. MRI of the pelvis was performed. Coronal postcontrast T1-weighted image shows a well-defined rounded hyperenhancing mass (arrow) along the expected course of the femoral nerve. The mass was excised, and pathologic analysis demonstrated a schwannoma.



Figure 22. Paraganglioma as a mimic of CD in a 28-year-old man with intermittent episodes of anxiety, nausea, feeling unwell, and profuse sweating. Axial contrast-enhanced CT image of the abdomen shows a heterogeneously hyperenhancing retroperitoneal mass (arrow) that partially encases the aorta without associated stenosis or occlusion. Resection demonstrated a paraganglioma.

the patients studied had a low incidence of associated TA-FRO), 89% for HHV8-MCD patients with HIV negativity, and 65% for HHV8-MCD patients with HIV positivity (1). Overall, the prognosis for HHV8-MCD has improved dramatically after the introduction of rituximab-based therapy.

Conclusion

CD is a complex and heterogeneous group of lymphoproliferative disorders that can manifest with varying clinical, imaging, and histopathologic features. UCD typically manifests as an incidental solitary enlarging mass, whereas MCD manifests acutely with systemic manifestations. A multidisciplinary approach to CD including clinical information, laboratory evaluation, and imaging workup is necessary to select appropriate regions for biopsy, exclude potential mimics, differentiate UCD from MCD, arrive at an accurate diagnosis, and ultimately manage CD. Imaging can be used to monitor response to treatment and detect development of any complications such as lymphoma. Surgical excision is typically cu-

rative for UCD, while MCD requires systemic therapy with a more guarded prognosis.

Author affiliations.—From the Department of Radiology (M.A.P., A.D.T.A., L.W., P.J.N., S.K.V.), Department of Laboratory Medicine and Pathology (A.J.W.), and Department of Hematology-Oncology (A.D.), Mayo Clinic, 200 First St SW, Rochester, MN 55905; Department of Radiology, Mayo Clinic, Jacksonville, Fla (L.F.A., J.D.L.); and Department of Radiology, Mayo Clinic, Phoenix, Ariz (A.K.). Presented as an education exhibit at the 2022 RSNA Annual Meeting. Received December 11, 2022; revision requested March 9, 2023, and received March 25; accepted April 6. **Address correspondence to** S.K.V. (email: Venkatesh.sudhakar@mayo.edu).

Acknowledgment.—The authors acknowledge the assistance of Sonia Watson, PhD, in preparation of the manuscript.

Disclosures of conflicts of interest.—M.A.P. RSNA R3 Resident Representative, member of *RadioGraphics* Visual Abstract Team. A.D. Grants from Bristol Myers Squibb, Takeda, Janssen, and Pfizer; personal fees from Janssen; support from Alnylam. S.K.V. Royalties or licenses from Springer for textbook. All other authors, the editor, and the reviewers have disclosed no relevant relationships.

References

- Dispenzieri A, Fajgenbaum DC. Overview of Castleman disease. *Blood* 2020;135(16):1353–1364.
- Castleman B, Towne VW. Case records of the Massachusetts General Hospital: Case No. 40231. *N Engl J Med* 1954;250(23):1001–1005.
- Castleman B, Iverson L, Menendez VP. Localized mediastinal lymphnode hyperplasia resembling thymoma. *Cancer* 1956;9(4):822–830.
- Kligerman SJ, Auerbach A, Franks TJ, Galvin JR. Castleman Disease of the Thorax: Clinical, Radiologic, and Pathologic Correlation—From the Radiologic Pathology Archives. *RadioGraphics* 2016;36(5):1309–1332.
- Carbone A, Borok M, Damania B, et al. Castleman disease. *Nat Rev Dis Primers* 2021;7(1):84.
- Simpson D. Epidemiology of Castleman Disease. *Hematol Oncol Clin North Am* 2018;32(1):1–10.
- Soumerai JD, Sohani AR, Abramson JS. Diagnosis and management of Castleman disease. *Cancer Contr* 2014;21(4):266–278.
- Dossier A, Meignin V, Fieschi C, Boutboul D, Oksenhendler E, Galicier L. Human herpesvirus 8-related Castleman disease in the absence of HIV infection. *Clin Infect Dis* 2013;56(6):833–842.
- El-Osta HE, Kurzrock R. Castleman's disease: from basic mechanisms to molecular therapeutics. *Oncologist* 2011;16(4):497–511.
- Siegel MO, Ghafouri S, Ajmera R, Simon GL. Immune reconstitution inflammatory syndrome, human herpesvirus 8 viremia, and HIV-associated multicentric Castleman disease. *Int J Infect Dis* 2016;48:49–51.
- Bonekamp D, Horton KM, Hruban RH, Fishman EK. Castleman disease: the great mimic. *RadioGraphics* 2011;31(6):1793–1807.
- Luigetti M, Conte A, Bisogni G, Del Grande A, Sabatelli M. 'White nails': skin changes in POEMS syndrome. *Eur Neurol* 2015;73(1-2):89, 112.
- Dispenzieri A. POEMS syndrome: 2021 update on diagnosis, risk-stratification, and management. *Am J Hematol* 2021;96(7):872–888.
- Hill AJ, Tirumani SH, Rosenthal MH, et al. Multimodality imaging and clinical features in Castleman disease: single institute experience in 30 patients. *Br J Radiol* 2015;88(1049):20140670.
- Haap M, Wiefels J, Horger M, Hoyer A, Müssig K. Clinical, laboratory and imaging findings in Castleman's disease: the subtype decides. *Blood Rev* 2018;32(3):225–234.
- Chen YB, Rahemtullah A, Hochberg E. Primary effusion lymphoma. *Oncologist* 2007;12(5):569–576.
- Dupin N, Diss TL, Kellam P, et al. HHV-8 is associated with a plasmablastic variant of Castleman disease that is linked to HHV-8-positive plasmablastic lymphoma. *Blood* 2000;95(4):1406–1412.
- Hwang SO, Lee TH, Bae SH, et al. Transformation of Castleman's disease into follicular dendritic cell sarcoma, presenting as an asymptomatic intra-abdominal mass. *Korean J Gastroenterol* 2013;62(2):131–134.
- Cakir E, Aydin NE, Samdanci E, Karadag N, Sayin S, Kizilay A. Follicular dendritic cell sarcoma associated with hyaline-vascular Castleman's disease. *J Pak Med Assoc* 2013;63(3):393–395.
- Kafaei L, Millard T, Wotherspoon A, Attygalle AD, Cunningham D, Sharma B. Follicular dendritic cell sarcoma in the setting of Castleman disease. *Br J Haematol* 2022;197(5):512.
- Restrepo CS, Martinez S, Lemos JA, et al. Imaging manifestations of Kaposi sarcoma. *RadioGraphics* 2006;26(4):1169–1185.
- Addula D, Das CJ, Kundra V. Imaging of Kaposi sarcoma. *Abdom Radiol (NY)* 2021;46(11):5297–5306.
- Fajgenbaum DC, Uldrick TS, Bagg A, et al. International, evidence-based consensus diagnostic criteria for HHV-8-negative/idiopathic multicentric Castleman disease. *Blood* 2017;129(12):1646–1657.
- Yoshizaki K, Murayama S, Ito H, Koga T. The Role of Interleukin-6 in Castleman Disease. *Hematol Oncol Clin North Am* 2018;32(1):23–36.
- Fajgenbaum DC, Shilling D. Castleman Disease Pathogenesis. *Hematol Oncol Clin North Am* 2018;32(1):11–21.
- Yoshizaki K, Matsuda T, Nishimoto N, et al. Pathogenic significance of interleukin-6 (IL-6/BSF-2) in Castleman's disease. *Blood* 1989;74(4):1360–1367.
- Masaki K, Takaki S, Hyogo H, et al. Utility of controlled attenuation parameter measurement for assessing liver steatosis in Japanese patients with chronic liver diseases. *Hepatol Res* 2013;43(11):1182–1189.
- Lu K, Zhao Y, Xu W, Zhang C, Huang P. Ultrasound and radiological features of abdominal unicentric Castleman's disease: a case series study. *Medicine (Baltimore)* 2020;99(18):e20102.
- Ko SF, Hsieh MJ, Ng SH, et al. Imaging spectrum of Castleman's disease. *AJR Am J Roentgenol* 2004;182(3):769–775.
- Zhao S, Wan Y, Huang Z, Song B, Yu J. Imaging and clinical features of Castleman disease. *Cancer Imaging* 2019;19(1):53.
- Oksenhendler E, Boutboul D, Fajgenbaum D, et al. The full spectrum of Castleman disease: 273 patients studied over 20 years. *Br J Haematol* 2018;180(2):206–216.
- Zimmermann A. Hepatobiliary Castleman's Disease. In: *Tumors and Tumor-like Lesions in the Hepatobiliary Tract*. Cham, Switzerland: Springer, 2017; 1729–1744.
- Wang H, Wiecek RL, Zenilman ME, Desoto-Lapaix F, Ghosh BC, Bowne WB. Castleman's disease in the head of the pancreas: report of a rare clinical entity and current perspective on diagnosis, treatment, and outcome. *World J Surg Oncol* 2007;5(1):133.
- Wang J, Wang B, Chen DF. A rare submucosal tumor of the stomach. *Gastroenterology* 2013;144(4):e5–e6.
- Ergul E, Korukluoglu B, Yalcin S, Ozgun YM, Kusdemir A. Castleman's disease of the duodenum. *J Pak Med Assoc* 2008;58(12):704–706.
- Li F, Xiao L, Cai H, Li L. Colonic Castleman Disease on FDG PET/CT. *Clin Nucl Med* 2023;48(1):71–72.
- Koukalová R, Selingerová I, Řehák Z, Adam Z, Szturp P. FDG-PET/CT for initial staging and response assessment in Castleman disease: retrospective single-center study of 29 cases. *Klin Onkol* 2021;34(2):120–127.
- Jiang Y, Hou G, Zhu Z, Huo L, Li F, Cheng W. 18F-FDG PET/CT imaging features of patients with multicentric Castleman disease. *Nucl Med Commun* 2021;42(7):833–838.
- Lee ES, Paeng JC, Park CM, et al. Metabolic characteristics of Castleman disease on 18F-FDG PET in relation to clinical implication. *Clin Nucl Med* 2013;38(5):339–342.
- Pan Q, Li J, Li F, Zhou D, Zhu Z. Characterizing POEMS syndrome with 18F-FDG PET/CT. *J Nucl Med* 2015;56(9):1334–1337.
- Mao S, Dong J, Wang Y, Zhang C, Dong A, Shen J. Follicular Dendritic Cell Sarcomas: CT and MRI Findings in 20 Patients. *AJR Am J Roentgenol* 2021;216(3):835–843.
- Nahi H, Genell A, Wälinder G, et al. Incidence, characteristics, and outcome of solitary plasmacytoma and plasma cell leukemia: population-based data from the Swedish Myeloma Register. *Eur J Haematol* 2017;99(3):216–222.
- Nishino M, Ashiku SK, Kocher ON, Thurer RL, Boisselle PM, Hatabu H. The thymus: a comprehensive review. *RadioGraphics* 2006;26(2):335–348.
- Drevelgas A, Palladas P, Scordalaki A. Mediastinal germ cell tumors: a radiologic-pathologic review. *Eur Radiol* 2001;11(10):1925–1932.
- Caers J, Paiva B, Zamagni E, et al. Diagnosis, treatment, and response assessment in solitary plasmacytoma: updated recommendations from a European expert panel. *J Hematol Oncol* 2018;11(1):10.
- Figueiredo G, O'Shea A, Neville GM, Lee SI. Rare Mesenchymal Tumors of the Pelvis: Imaging and Pathologic Correlation. *RadioGraphics* 2022;42(1):143–158.
- Baez JC, Jagannathan JP, Krajewski K, et al. Pheochromocytoma and paraganglioma: imaging characteristics. *Cancer Imaging* 2012;12(1):153–162.
- Pilavaki M, Chourmouzi D, Kiziridou A, Skordalaki A, Zampoukas T, Drevelengas A. Imaging of peripheral nerve sheath tumors with pathologic correlation: pictorial review. *Eur J Radiol* 2004;52(3):229–239.
- Khanna L, Prasad SR, Sunnapwar A, et al. Pancreatic Neuroendocrine Neoplasms: 2020 Update on Pathologic and Imaging Findings and Classification. *RadioGraphics* 2020;40(5):1240–1262.